Is Pocketbook Voting Sensitive to Policy?[†]

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People often vote with their pocketbooks, but is this an effective accountability mechanism? Whether voters respond to the policy-induced component of their income changes or simply react to all changes regardless of cause remains unresolved: single-policy studies cannot distinguish the two, and research on pocketbook attribution relies on potentially biased self-reports. I introduce an approach linking a survey panel to policy microsimulation models, decomposing respondents' total income changes into policy-induced and non-policy components. This directly tests what component of income variation drives pocketbook voting. Applying this to the UK in the 2010s, where austerity produced substantial policy-driven income variation, I find that voters do not hold incumbents more accountable for policy-induced income changes. Instead, they respond indiscriminately to total income changes — a poor proxy for the effects of policy. Pocketbook voting is thus a weaker mechanism for policy accountability than commonly assumed.

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"... we did a lot for working people [but] we didn't tell people about it. We thought: just by legislating, people would know about it. Well, they don't!"

– Chuck Schumer, 'The Daily', 16 March 2025

How do voters hold democratic politicians accountable in a complex world? One widely documented shortcut is pocketbook voting: rewarding incumbents when personal finances improve and punishing them when they decline (Bechtel and Liesch 2020; Healy, Persson and Snowberg 2017; Tilley, Neundorf and Hobolt 2018). By tracking personal income changes, voters can motivate the government to make policies they like without closely following politics or analyzing macroeconomic trends. This form of "rough justice" (Fiorina 1981, p.4) promises to discipline politicians by tying their survival to citizens' economic well-being.

Yet scholars disagree about which income changes voters respond to. Do they hold incumbents accountable for all income changes, regardless of their cause, as Fiorina (1981) famously argued? Or do they only respond to income changes that are 'policy-induced', i.e. caused by policy changes such as tax reforms, new spending programs, or changes to the welfare system? The first model – outcome-based pocketbook voting – is undoubtedly easier for voters who need only "calculate changes in their own welfare" (Fiorina 1981, p. 5) to decide the incumbent's fate. However, as Kramer (1983) and other scholars have pointed out, personal income changes are "contaminated" (p. 99) by idiosyncratic and exogenous variation unrelated to government policy, like workplace changes, life-cycle transitions, family circumstances, and health shocks. As a result, under outcome-based pocketbook voting, electoral sanctions may largely reflect income variation that is only weakly related to the incumbent's policy choices. The second model – policy-based pocketbook voting – involves the more demanding task of attributing responsibility for changes in economic outcomes, but results in a tighter link between policy effects and electoral sanctions (Achen and Bartels 2016; Duch and Stevenson 2008; Powell and Whitten 1993).

While recent research appears to support the policy-based model, existing research has not definitively resolved which model of pocketbook voting prevails. One strand of the economic voting literature examines whether voters are more responsive to income changes that they sub-

jectively attribute to government policy than to other income changes (e.g. Tilley, Neundorf and Hobolt 2018). However, these studies rely on self-reported attributions without objective benchmarks, leaving open the possibility of rationalized or biased attribution judgments (Feldman 1982; Larsen 2021). While citizens aspire to policy-based pocketbook voting, these studies do not show whether they achieve it.

Another strand – the policy feedback literature – focuses on actual behavioral responses to policies (e.g. Kogan 2021; Manacorda, Miguel and Vigorito 2011; Pop-Eleches 2012). But these designs lack a crucial counterfactual: how voters would have responded to comparable income changes had they not resulted from policy. Evidence that voters respond to policy-induced income shocks is thus equally consistent with outcome-based pocketbook voting, as it may simply reflect voters sanctioning incumbents for any income changes regardless of their cause. Moreover, most of this work focuses on single cases of highly salient programs (Kogan 2021), which are somewhat atypical fiscal events and tell us little about how reliably voters respond to policy changes in general. In sum, there is little direct evidence on whether policy-induced or total income changes drive pocketbook voting.

This paper adjudicates between the two models of pocketbook voting with a novel empirical approach. Moving beyond singular policies, I examine voters' responses to the net effects of all government policies that directly affect their disposable incomes using the advanced policy-microsimulation model, UKMOD. UKMOD is essentially an advanced disposable income calculator that can apply any historical policy system to any set of individual characteristics, enabling counterfactual analysis. This allows me to decompose changes in the disposable incomes of around 30,000 surveyed British voters over a decade into (i) the net effect of changes to policies that directly affect incomes (i.e. taxes and transfers), and (ii) other income fluctuations caused by e.g. changes in personal circumstances. I can thus capture how reliably voters respond to such 'policy-induced' income changes on average as well as any differential responsiveness to such changes compared to other income changes. This provides a rare test of whether pocketbook voting is mostly policy-based or outcome-based.

Using this approach, I find limited support for policy-based pocketbook voting. With conventional survey measures of incumbent party support as outcomes, I find that voters sanction incumbents no more for policy-induced income changes than other income changes, instead responding equally to all changes regardless of their origin. Even the most optimistic estimates rule out voters weighting policy-induced changes more than about 2 times as heavily as other changes, which is far below what policy-sensitive pocketbook voting would require given the predominance of non-policy variation. Nor is there strong evidence that voters respond more to changes in taxes paid or benefits received than to market income changes. Subgroup analyses confirm that this pattern holds across the electorate, with no group displaying strong signs of policy-based pocketbook voting. This would be consistent with voters being less able to notice policy-induced income changes the more their income changes for other reasons, following a signal-extraction logic. These findings suggest that voters do not reliably hold incumbents accountable for the direct income effects of their policies.

The income decomposition further reveals that these policy-induced changes account for roughly a fifth of total income variation on average. Thus, when voters respond equally to all income changes, pocketbook voting is primarily driven by other income changes. Most pocketbook voting therefore represents "rough justice" based on total income changes, which bear little relation to the direct income effects of government policy.

These findings have important implications for democratic accountability and party strategy. If voters do not reliably sanction incumbents for policy-induced income changes, parties may struggle to secure electoral support simply by redistributing resources, casting doubt on strategies like "deliverism" (Bhargava, Shams and Hanbury 2024). The normative implications are ambiguous: while the disciplining power of pocketbook voting may be weaker than assumed, low policy-sensitivity may also reduce incumbents' incentives for tactical redistribution. I discuss these implications in the concluding section.

Pocketbook Voting and Policy-Sensitivity

Recent studies suggest that pocketbook voting significantly shapes electoral behavior, influencing the electoral rewards and punishments incumbents can expect from their policies (Bechtel and Liesch 2020; Healy, Persson and Snowberg 2017; Tilley, Neundorf and Hobolt 2018). Yet, the relationship between pocketbook voting and incumbent incentives is not straightforward. When voters reward and punish incumbents for personal income changes, they incentivize incumbents to raise voters' incomes, especially for electorally pivotal groups. But for this mechanism to shape incumbent incentives, incumbents must expect to be rewarded specifically for policies that benefit voters and punished for those that harm them. Strong pocketbook voting alone does not guarantee this outcome because voters' incomes change for many reasons besides policy, like workplace changes, family circumstances, and life-cycle transitions. If voters respond equally to all income changes regardless of their cause, they may inadvertently punish incumbents whose policies actually helped them, or reward those whose policies hurt them. Under these conditions, even strong pocketbook voting may fail to provide clear policy incentives.[1] Whether pocketbook voting translates into strong policy incentives depends, I argue, on two features of voters' economic circumstances and behavior: how much weight voters place on policy-induced versus other income changes when they vote, and how large policy-induced income changes are relative to other sources of income variation.

A Model of Pocketbook Voting

Consider a simple pocketbook voting model of electoral accountability. In it, incumbent policies have a net effect on each voter's pocketbook that is either positive or negative. Voters, in turn, can choose to support or reject the incumbent at the ballot box (Ferejohn 1986; Key 1966). Following Kramer (1983), the net change in i's disposable income in period t can be defined as a sum of two jointly exhaustive components: policy-induced income changes, e.g. due to changes to tax brackets

[1] To be sure, one cannot draw too determinate conclusions about incumbent incentives based on individual-level voter behavior alone (Ashworth and de Mesquita 2014). This argument therefore avoids deriving precise implications for incumbent behavior. I return to this point in the discussion.

or new government benefits, and non-policy income changes, e.g. due to idiosyncratic changes or broader exogenous shocks. Ignoring other non-pocketbook considerations in vote choice for simplicity[2], we can model i's binary incumbent vote choice as a function of these components as follows:

$$\text{incumbent vote}_{it} = \beta_p \Delta y_{it}^{policy} + \beta_{np} \Delta y_{it}^{nonpolicy} \tag{1}$$

where the sum of Δy_{it}^{policy} and $\Delta y_{it}^{nonpolicy}$ is the total change in i's disposable income, and the β's are the weights voters put on each type of income change in deciding their vote.

Based on this equation, I define an electorate's *policy-sensitivity* as the share of voters whose incumbent vote aligns with the sign on Δy_{it}^{policy} in a given election.[3] The policy-sensitivity of the electorate is what determines the strength of incumbent incentives: the more reliably voters sanction the incumbent in line with their personal economic gain or loss from their policies, the more reliably incumbents can expect electoral rewards for the income effects of their policies. For simplicity, I refer to cases where voters reward (punish) incumbents whose policies increased (decreased) their income as 'concordant' sanctions, and the opposite as 'discordant' sanctions. What, then, determines whether voters' sanctions are mostly concordant or discordant? The model suggests two key factors: how much weight voters place on policy-induced versus other income changes ('attribution'), and how much of total income variation stems directly from policy versus other sources (the magnitude of non-policy variation).

The first and most obvious way for sanctions to be concordant in this model is if voters act just on policy-induced income changes ($\beta_p > 0$) and ignore non-policy changes ($\beta_{np} = 0$). This is the policy-based model of pocketbook voting espoused by Kramer (1983) in his influential article arguing that pocketbook voting can yield strong and reliable electoral sanctions even if the overall relationship between income changes and incumbent support is weak. Other scholars are likewise optimistic that voters are able to judge when policy is responsible for pocketbook changes

[2]One can think of these as an added term, α_{it}, in Equation 1 representing the sum weight of alternative considerations.

[3]Equivalently, an *individual's* policy-sensitivity is this share within that individual across elections over time. I mostly focus on electorate-level policy-sensitivity because it is more intuitive.

and when it is not (Ashworth, de Mesquita and Friedenberg 2018; Fowler and Hall 2018; Fowler and Montagnes 2015, 2023; Tilley, Neundorf and Hobolt 2018). Placing disproportionate weight on policy-induced income changes is commonly called 'attribution', a term that suggests voters psychologically identify which income changes are caused by policy and filter out the rest. There is reason to think such attribution is feasible: voters can directly observe changes in the taxes or contributions they pay or benefits they receive, and media coverage can alert them to policies that affect their incomes (Ciobanu 2024; Mutz 1994). But attribution need not require deliberate identification of policy-induced changes. If such changes are naturally more salient, or if voters learn from more politically attentive peers (Lupia and McCubbins 1998), the same behavioral pattern can arise through less cognitively demanding channels.

Yet, voters may not place disproportionate weight on policy-induced income changes at all. By contrast to Kramer's model, Fiorina (1981)'s seminal theory of pocketbook voting posits no such attribution.[4] In Fiorina's outcome-based model, voters exert "rough justice" (p.4) based on total income changes (i.e. $\beta_p = \beta_{np}$). This pattern can arise for two quite different reasons. Many citizens may lack the financial literacy to parse which income changes stem from policy (Abeler and Jäger 2015; Bhargava and Manoli 2015; Peters 1991) and many scholars doubt that ordinary voters can reliably do so (Achen and Bartels 2016; Feldman 1984; Gomez and Wilson 2001; Healy, Malhotra and Mo 2010). But such outcome-based voting may also reflect a rational heuristic rather than a cognitive shortcoming. Following the logic of classical principal-agent models with unobserved agent effort, the principal may rationally sanction the agent for total outcomes when the 'effort signal' is unavailable or too costly to extract (Holmström 1979). Under either reading, the behavioral prediction is the same: voters respond to total income changes without placing disproportionate weight on the policy-induced component.

Policy-based pocketbook voting is, by definition, policy-sensitive. But what about outcome-based pocketbook voting? Some scholars argue that without attribution, policy-sensitivity essen-

[4]To be sure, while Fiorina's model is fundamentally outcome-based, he does not fully ignore attribution. For instance, he interprets partisan interactions with pocketbook effects as reflecting differences in attribution (p. 112).

tially collapses (Achen and Bartels 2016; Kramer 1983; Sances 2017; Wolfers 2002), while others disagree (Ferejohn 1986; Fiorina 1981; Key 1966; Miller 2013). The answer turns on a second factor: the magnitude of non-policy income variation, $\Delta y_{it}^{nonpolicy}$, relative to Δy_{it}^{policy}. When non-policy variation is small, total income changes largely reflect the effects of policy, and voters' sanctions will tend to be concordant even under Fiorina's model – in the extreme case of $\Delta y_{it}^{nonpolicy} = 0$, perfectly so. The disagreement thus turns on an empirical question: how large is non-policy income variation relative to the policy-induced component? The reason Kramer and others are sceptical of policy-sensitivity without attribution is their presumption that policy-induced income changes are relatively small. Thus, Kramer, for instance, makes an educated guess that income changes directly caused by policy make up 0-10% of total income variation on average (fn. 13, 1983).

Figure 1 illustrates these relationships using simulated electorates of pure pocketbook voters that vary in their degree of attribution ($\beta_p : \beta_{np}$), as shown by the five curves, and the share of income variation caused by policy on the horizontal axis (see Appendix A for details).

Under Kramer's policy-based model, voters' sanctions always align with their policy-induced income changes regardless of the level of non-policy variation. Fiorina's outcome-based pocketbook voting performs about as well when non-policy variation is limited. But as the non-policy component increases in magnitude, the two models quickly diverge. At Kramer's estimated level of over 90% non-policy variation, even substantial attribution – such as weighting policy-induced changes five or ten times as heavily as other changes – will result in sanctions that align with the sign of the policy-induced component only modestly more often than chance. Still, as the darker curves with varying degrees of attribution show, voters need not engage in perfect attribution for their sanctions to largely align with their policy-induced income changes in many scenarios.

Summing up, the strength of pocketbook voting itself tells us little about its power to shape incumbent incentives. We must further know whether voters 'attribute' income changes, putting more weight on changes that are policy-induced (policy-based pocketbook voting), or whether they just respond to total income changes (outcome-based pocketbook voting). In the latter case, policy-sensitivity hinges on how reliably total income changes capture the effects of policy rather

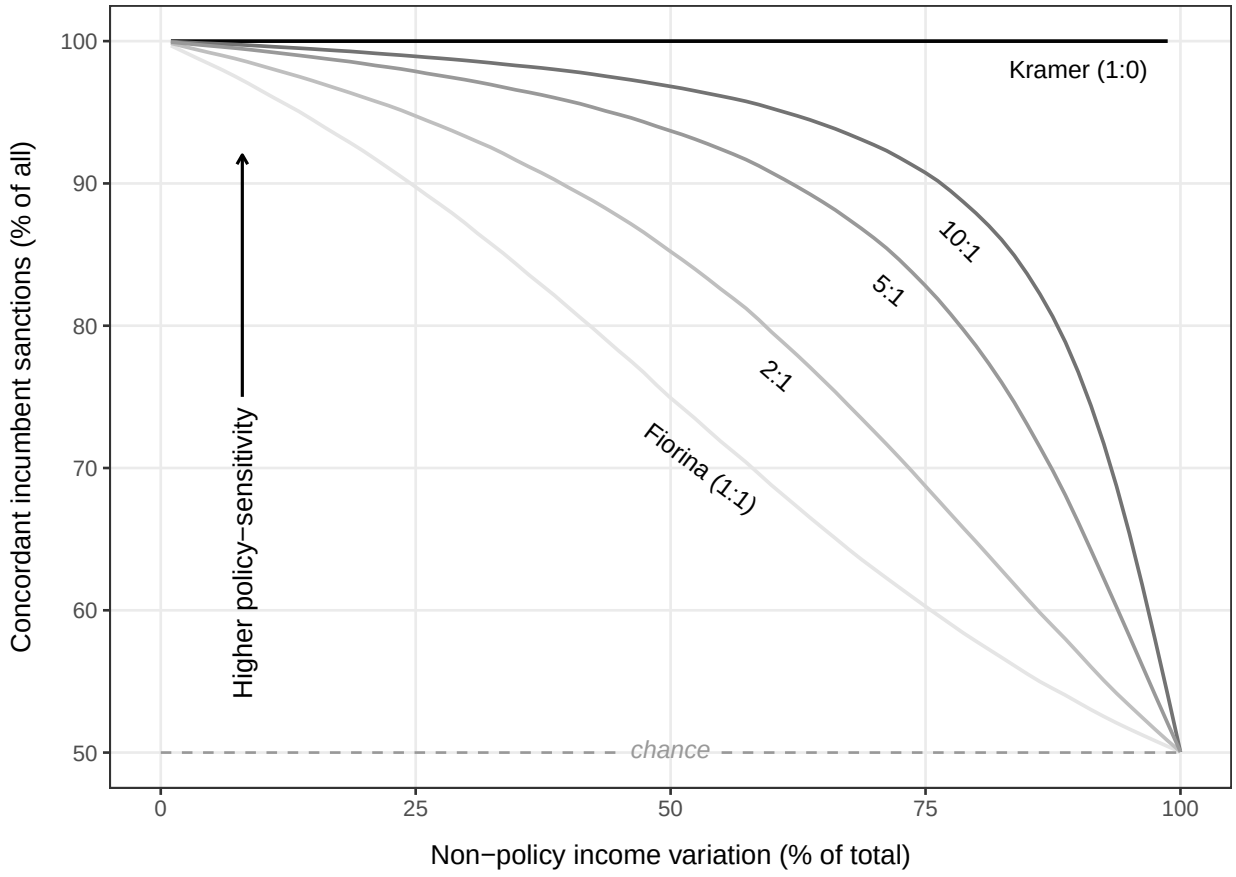


Figure 1: Policy-sensitivity across simulated electorates of pure pocketbook voters. Each electorate consists entirely of pocketbook voters with different combinations of attribution ($\beta_p : \beta_{np}$), as shown by gray curves ranging from none (1 : 1) to perfect (1 : 0), and magnitudes of non-policy variation as indicated by the horizontal axis ranging from 0% to 100%. The vertical axis indicates the share of concordant incumbent sanctions achieved by each electorate going from chance (50%) to perfectly concordant (100%). See Appendix A for technical details on the simulations.

than non-policy income variation. Standard models of pocketbook accountability must therefore assume either a high degree of pocketbook attribution or a relatively small non-policy component. While the relative magnitudes of the policy-induced and non-policy components has not been the subject of empirical research, much research has centered on attribution. I turn to the existing evidence on attribution next.

Empirical Challenges to Inference in Existing Research

Whether voters distinguish government-caused from other economic outcomes has been a lively topic since Fiorina (1981) and Kramer (1983)'s seminal work. Much of this evidence has been read as supporting policy-based pocketbook voting, yet it is indirect and cannot directly adjudicate between the two models.

First, one strand of research on pocketbook voting finds that voters seem more responsive to income changes that they themselves attribute to government action (Abramowitz, Lanoue and Ramesh 1988; Feldman 1982; Lau and Sears 1981; Marsh and Tilley 2010; Tilley, Neundorf and Hobolt 2018; although see Huber, Hill and Lenz 2012). In the perhaps most direct test of the question, Tilley, Neundorf and Hobolt (2018) find that income changes "have a greater impact when they can be [...] attributed directly to government policies, such as welfare payments and other government transfers" (Tilley, Neundorf and Hobolt 2018, p. 567). However, these self-reported attributions may be "tainted by rationalization" (Kiewiet and Rivers 1984, p. 383) or self-serving biases (Larsen 2021). Without objective benchmarks, we cannot assess whether voters actually get attribution right. Indeed, other studies argue that voters are overly reluctant to link finances to politics, attributing only a very small share of their income to government (Brody and Sniderman 1977; Conover 1984; Feldman 1984). Thus, while this strand shows that *subjective* attributions predict vote choice, it cannot establish whether voters actually place greater weight on policy-induced income changes in their voting behavior.

The second category of evidence comes from studies of policy feedback. These studies show voters sometimes reward incumbents for targeted benefits (Kogan 2021; Manacorda, Miguel and Vigorito 2011; Pop-Eleches 2012; Rendleman and Yoder 2024; Zucco Jr. 2013), especially when policies are highly visible, salient, publicized and traceable to government (Hamel 2024; Mettler 2011; Shanks-Booth and Mettler 2019; Soss and Schram 2007). But these designs lack a key counterfactual: how would voters respond to comparable income changes not caused by policy? Indeed, research on 'blind retrospection' finds that voters sometimes sanction incumbents for shocks that cannot be tied directly to policy, i.e. the pattern predicted by outcome-based pocketbook vot-

ing (Achen and Bartels 2016; Healy and Malhotra 2010; Healy, Malhotra and Mo 2010; Wolfers 2002; although see Fowler and Montagnes 2015, 2023; Graham et al. 2023). Without observing responses to both policy-induced and non-policy income changes, these designs cannot adjudicate between the two models. Moreover, these studies have mostly focused on rollouts of high-profile public spending programs (Kogan 2021) which are somewhat atypical fiscal events. It is therefore unclear how much can be inferred about the bulk of pocketbook variation in voters' daily lives, including the income effects of less visible policies like reforms of tax brackets or benefit eligibility criteria. Evidence of policy feedback in specific cases is therefore consistent with attribution being rare overall.

Finally, the literature on clarity of responsibility finds that voters sanction incumbents more heavily for economic conditions when institutional arrangements and government characteristics make incumbent responsibility clearer (Duch and Stevenson 2008; Hobolt, Tilley and Banducci 2013; Powell and Whitten 1993). While these designs sidestep subjective attributions by exploiting cross-country variation in institutional clarity, they cannot rule out country-level confounders that covary with these same institutional characteristics (for an exception, see Larsen 2019). The evidence for policy-based pocketbook voting is therefore very indirect.

In sum, none of these designs directly tests whether voters weight policy-induced income changes more heavily than other changes in their voting behavior. Doing so requires measuring both components at the individual level and comparing their electoral effects. In the next section, I describe an approach that achieves this, using microsimulation models to separately estimate each voter's policy-induced and non-policy income changes. Combined with panel data on vote intentions, this permits a direct test of which model better describes pocketbook voting. It additionally provides an estimate of how large the policy-induced component actually is relative to non-policy variation; a quantity that, as Figure 1 shows, determines how reliably outcome-based pocketbook voting produces concordant sanctions.

Methods and Data

My empirical approach involves first, estimating the share of policy-induced income changes in voters' pocketbooks, and second examining how reliably they sanction incumbents for them. For this, I turn to the policy microsimulation model, UKMOD. A microsimulation model is a computational model that applies a set of pre-defined tax and benefit rules to individual-level data on market incomes and demographics to calculate post-policy disposable incomes. Developed by a team of University of Essex economists, UKMOD and its relatives are widely used for policy analysis by governments and researchers and modeled incomes have been extensively validated in economics (see Richiardi, Collado and Popova 2021). Despite this, they have barely been used in political science (Avram and Popova 2022; Elkjær and Mushövel 2023).

I exploit UKMOD to estimate the policy-sensitivity of pocketbook voting in three steps. First, I apply UKMOD to a large panel dataset, the UK Household Longitudinal Survey (UKHLS), to calculate respondents' disposable incomes under actual and counterfactual policy systems. Second, I use Bargain and Callan (2010)'s decomposition approach extended by Paulus and Tasseva (2020) to separate each respondent's yearly income change into a policy-induced and a non-policy component. The decomposition is implemented using UKMOD's Policy Effects Tool, which automates the counterfactual calculation. Third, I regress incumbent support on both components to test whether voters sanction incumbents more reliably for policy-induced income changes than for other income changes.

The microsimulation decomposition

UKMOD calculates disposable incomes based on information about individuals and a policy system. Specifically, it applies a policy system from a given year, s_t, to a sample of individuals with characteristics, c_{it}, to compute their disposable incomes, y_{it}:

$$y_{it} = s_t(c_{it})$$

Here, the individual's characteristics, c_{it}, comprise market incomes and policy-relevant demo-

graphics (number of children, employment status, age, and the like). The policy system, s_t, encodes the tax and benefit rules in force in a given year together with their monetary parameters (CPI-indexed to a common price year when comparing across years; see below).

The key feature of this setup is that UKMOD can re-run the calculation with a *different* policy system, holding the person constant. That is, one can ask: what would this individual's disposable income have been if last year's rules still applied? By comparing this counterfactual income to the individual's actual income under current rules, the model isolates the net effect of policy changes on that person's disposable income. This is what makes the decomposition possible.

To formalize this, note that the total change in i's disposable income between an initial period $t = 0$ and a final period $t = 1$ can be written as:

$$\Delta y_{i,0\to1}^{total} = s_1(c_{i,1}) - s_0(c_{i,0})$$

This total change reflects two simultaneous shifts: changes in the policy system, $s_0 \to s_1$, and changes in the individual's market incomes and demographics, $c_{i0} \to c_{i1}$. These are confounded in observational data. The decomposition separates them by computing the counterfactual quantity $s_0(c_{i,1})$ – what i's income would have been under the old policy system given their current characteristics.[5] Following Bargain and Callan (2010)'s approach, the total change can then be decomposed as:

$$\Delta y_{i,0\to1}^{total} = \underbrace{\{s_1(c_{i,1}) - s_0(c_{i,1})\}}_{\Delta y_{i,0\to1}^{policy}} + \underbrace{\{s_0(c_{i,1}) - s_0(c_{i,0})\}}_{\Delta y_{i,0\to1}^{nonpolicy}}$$

All changes are expressed in real, CPI-adjusted terms: I apply CPI indexation factors that adjust the earlier system's monetary parameters to the later year's prices, so that s_0 in the decomposition above denotes the CPI-indexed version of the earlier policy system. This ensures that routine

[5]This is what Bargain and Callan (2010) refers to as "end-weighting". Mathematically, the policy-induced income change could also be decomposed with respect to the counterfactual quantity $s_1(c_0)$. Psychologically, this would be equivalent to asking "given who I was *before* the reform, how does the new system compare to the old?" which is a more convoluted retrospective evaluation than what retrospective voting usually theorizes.

inflation-matching adjustments to policy parameters are absorbed into the non-policy component and only real changes in generosity register as policy-induced. See Appendix N for technical details on the indexation procedure and a robustness check using no indexation.

What, then, does the policy-induced component capture in concrete terms? It registers changes in the policy rules that determine what voters pay to or receive from the state: altered tax rates, changed social contribution payments, new benefit formulas, changed eligibility thresholds. One of the most visible UK examples during this period was the near-doubling of the income tax personal allowance, from £6,475 in 2010–2011 to £12,500 by 2019–2020. For a basic-rate taxpayer earning, say, £25,000, this single reform reduced their annual tax bill by roughly £1,200 over the decade. The non-policy component captures the entire net effect – the £10,000 earnings gain and the £2,000 tax increase alike. Failures to uprate benefits in real terms also register as policy-induced changes. When the government froze most working-age benefits in cash terms from 2016 to 2020, recipients experienced no nominal change in their benefit rates, but because the decomposition operates in real terms, the freeze registers as a policy-induced reduction in income. The government chose not to maintain the real value of benefits, and the model records that choice.

The non-policy component captures everything driven by changes in personal circumstances: job changes, pay rises, retirement, shifts in household composition. Suppose a worker earns £30,000 in one year and receives a promotion to £40,000 the next. They now pay more income tax – roughly £2,000 more per year at the basic rate. That additional tax is real, and the worker likely notices it, but it is not a policy-induced change: the tax rules did not change, only the worker's circumstances did. In additional analyses reported in the appendices, I test whether voters respond more crudely to such overall changes in what they pay in taxes or contributions or what they receive in benefits, regardless of their causal origin.

I implement the decomposition with respect to *yearly* changes in disposable income. The UK fiscal policy system is a largely unified, national system and changes to taxes and benefits are generally implemented once a year, in April, and announced only a couple of weeks before[6] (Reis

[6]With a few exceptions towards the end of the covered time period; see Appendix B for an overview.

and Tasseva 2020). UKMOD tracks these fiscal years such that UKMOD's 2010 policy system was in force from April 2010 to March 2011 and so on (Reis and Tasseva 2020). Since changes to the system are discrete rather than continuous in time, survey panel respondents can be divided into fiscal year windows within which everyone was subject to the same policy system (see Appendix B). In addition, yearly income changes correspond closely to the time-frame on which pocketbook voting is usually studied and assumed to operate.

What the decomposition includes and excludes

While the concept of policy-sensitivity refers broadly to any effects of government policy on voters' incomes, my operationalization focuses on policies that directly regulate what voters receive from or pay to the government: taxes, tax credits, social contributions, and transfers. These instruments dominate both classical theories of distributive politics and the empirical literature on policy feedback (e.g., Mettler 2011; Pop-Eleches 2012; Rendleman and Yoder 2024; Sances 2017; Zucco Jr. 2013), and they provide the strongest test of policy-sensitivity. If voters are not responsive to these most salient instruments of redistribution, the scope for meaningful policy-sensitivity is likely limited overall. This focus entails three scope boundaries.

First, as UKMOD is a *static* microsimulation model, the policy-induced component captures only the mechanical, 'morning-after' effect of policy changes on income, holding behavior constant. This excludes indirect and macroeconomic effects of fiscal policy, but for the purpose of constructing a treatment variable this is desirable: it provides a measure of exogenous policy exposure uncontaminated by endogenous behavioral responses. Second, a few instruments (principally state pensions and disability benefits) are not simulated from legislative rules in UKMOD and instead enter at reported survey values. Because these cannot support the counterfactual experiment, I exclude them from all income components so that the decomposition is not contaminated by variation that cannot be cleanly attributed (see Appendix C for coverage details). Third, the model calculates entitlements and liabilities rather than actual take-up or compliance, so I technically estimate intent-to-treat effects. If non-take-up is stable within individuals over time, the panel design captures the correct change in entitlement even when the level is overstated. Further, from the

perspective of government accountability, the relevant quantity is arguably what citizens stand to gain or lose under the law.

Data and validation

UKMOD is extensively documented and validated against official data, and it closely emulates the policy simulation models used by UK government departments, sharing similar scope, assumptions, and results (Bronka, Popova and Richiardi 2023; Richiardi, Collado and Popova 2021).[7] I apply UKMOD to the UK Household Longitudinal Survey (UKHLS), an annual survey of a nationally representative sample of around 40,000 British households.[8] I use data from 10 yearly waves covering fiscal years 2010–2019 (Reis and Tasseva 2020).[9] The panel is pre-linked to UKMOD by other researchers, who match its granular income variables and demographics to UKMOD's policy functions (Bronka, Popova and Richiardi 2023). The fact that benefits, taxes and contributions are calculated from legislative rules generally provides higher precision than survey self-reports, where there is e.g. a tendency for benefits to be underreported (Bronka, Popova and Richiardi 2023, p. 6).

Respondents' market incomes are derived from self-reported incomes in the UKHLS panel. While not as precise as administrative income records (Healy, Persson and Snowberg 2017), the UKHLS income data are unusually high in quality and granularity, and have been used in prior studies of pocketbook voting (Tilley, Neundorf and Hobolt 2018). The dataset comprehensively captures income sources, including standard wages, earnings from casual jobs, private pensions, and passive income from investments and properties (Bronka, Popova and Richiardi 2023, p. 22). When benchmarked against official statistics, market incomes in the UKHLS align closely, with

[7]UKMOD is a post-Brexit offshoot of the EUROMOD microsimulation model, customized for the UK context. It includes several improvements but remains largely identical in structure and function. See Richiardi, Collado and Popova (2021) for details.

[8]The UKHLS is based on a proportionately stratified, equal probability sample of residential addresses and includes booster samples to ensure representativity of immigrant and ethnic minority groups (Bronka, Popova and Richiardi 2023). Interviews are conducted physically or by phone for all waves, except for an online subsample in the final two waves. My sample is restricted to respondents of voting age who are members of fully-responding households.

[9]The Policy Effects Tool splits its calculations in such a way that total income changes cannot be calculated for 2015-2016. The year 2016 is therefore excluded from main analyses.

discrepancies largely confined to income under-reporting among top earners (Bronka, Popova and Richiardi 2023).

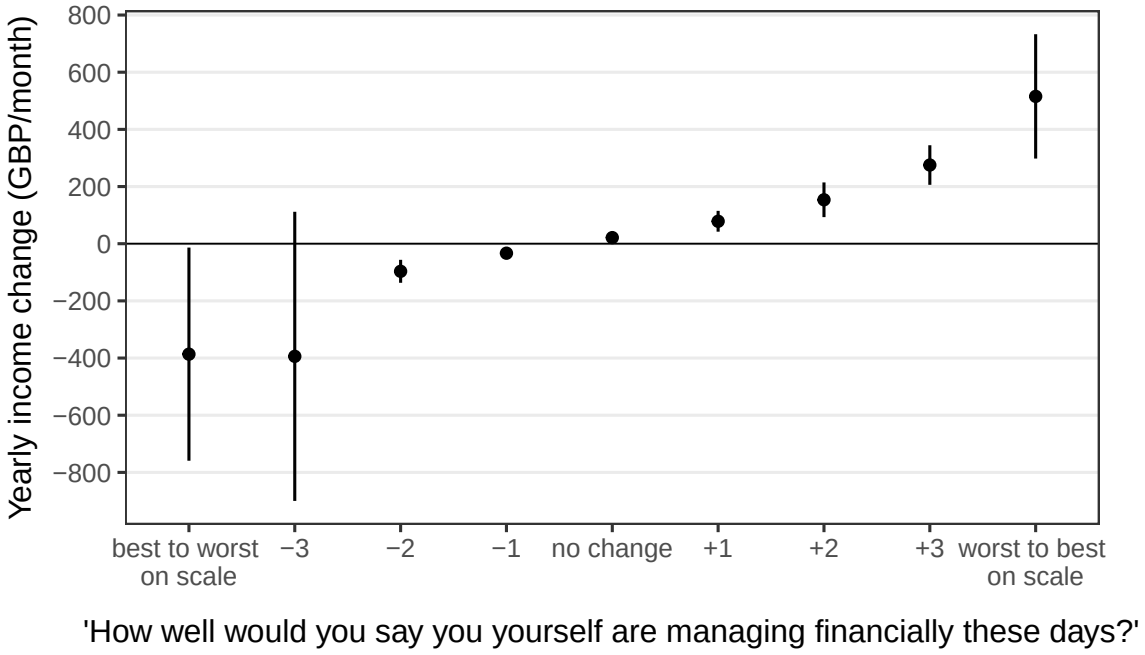


Figure 2: Average change in total disposable income by change in subjective financial evaluations across respondent-year observations.

To validate that the model's estimated disposable incomes reflect respondents' subjective experiences, I use a survey item from the UKHLS tapping respondents' evaluations of their current financial situation: "How well would you say you yourself are managing financially these days?" with five options ranging from "living comfortably" to "finding it very difficult". Do these subjective evaluations track disposable incomes as calculated by UKMOD? To validate the linked data, Figure 2 plots the relationship between changes in subjective pocketbook evaluations (horizontal axis) as measured in the UKHLS and actual disposable income changes as calculated by UKMOD. As shown, there is a tight relationship between calculated income changes and subjective evaluations. Respondents reporting the same pocketbook evaluation two years in a row see a near-zero change in their disposable income, while each step change on the scale is associated with a sub-

stantial and consistent change in disposable income. In a linear model, a one-point improvement on the scale (toward "living comfortably") is associated with a £65/month increase in disposable income on average ($SE = 6.07$, $p < 0.001$). This provides reassuring validation that UKMOD's calculated disposable incomes mirror respondents' subjective experience.

The case of the UK in the 2010's

I study policy-sensitivity in economic voting in the United Kingdom from 2010-2019. This is the maximum window for which UKMOD could be linked with the UKHLS. It is also a good case for studying economic voting in general and the policy-sensitivity of pocketbook voting in particular. By focusing on the UK, I can estimate the policy-sensitivity of pocketbook voting in a context where pocketbook voting itself is well-established with the same survey data (Tilley, Neundorf and Hobolt 2018). Further, the UK's fiscal policy system is relatively strongly redistributive (Hasell 2023) with substantial variation during the period of study (Elkjær and Mushövel 2023).

Further, the UK in the 2010's is a highly likely case for policy-based pocketbook voting. The Westminster system concentrates fiscal authority in the central government and typically produces single-party majorities with clear lines of accountability (Duch and Stevenson 2008; Powell and Whitten 1993). Apart from the 2010-2015 coalition with the Liberal Democrats, the Conservatives governed alone throughout the period.

The period also saw unusually large policy-induced income variation, driven in part by successive Conservative austerity programs. The universal credit overhaul affected millions of households while income tax rates and brackets changed repeatedly (see Appendix C for a year-by-year timeline). Importantly, despite austerity reducing aggregate spending, the distributional effects of tax and welfare reforms were heterogeneous: just over half of households gained overall — primarily from income tax cuts and the National Living Wage — while benefit freezes and cuts to tax credits concentrated losses among transfer-dependent households (Equality and Human Rights Commission 2018). The high public salience of these changes should, all else equal, make policy-induced income changes more 'traceable' and thus make subjective attribution easier (Hamel 2024).

Variables and estimation

Vote choice The main analysis estimates how voters sanction incumbents for disposable income changes depending on whether they are policy-induced or not. To measure support for the incumbent, I use a survey measure of vote intention for the government party from the UKHLS, coded as an indicator for incumbent party support and scaled to 0–100 so that regression coefficients can be read directly as percentage-point changes in the probability of incumbent support.[10] The resulting variable is widely used to measure vote intention in the British case and is strongly correlated with actual vote choice in election years (Tilley, Neundorf and Hobolt 2018).

Income variables I measure three types of disposable income variation: total income changes, policy-induced income changes, and non-policy income changes. All encode the individual-level income change from the last time the respondent took the survey (roughly a year prior) to the time of the interview. Specifically, UKMOD computes monthly disposable income as a snapshot at each survey point, and the policy-induced component reflects the instantaneous effect of the current fiscal year's rules on the respondent's income at that point — not cumulative exposure over the inter-interview window. Because each observation is assigned to the fiscal year in force at the interview date, and the counterfactual applies the previous fiscal year's rules to the same individual at the same point, the decomposition is unaffected by the proportion of the inter-interview window falling in different fiscal years.

I use two versions of each income change variable. To account for skewness, the main specifications use the inverse hyperbolic sine (IHS) transformation, which accommodates zero and negative values and approximates the log for large absolute values (Bellemare and Wichman 2020; McKenzie 2023). In addition, I use simple raw disposable income changes scaled to monthly income in £1,000 increments.

[10]The UKHLS measures party support through a sequence of three items: whether the respondent supports a particular party, whether they feel closer to one party than others, and which party they would vote for if there were an election tomorrow. Following Tilley, Neundorf and Hobolt (2018), the incumbent support variable is coded as an indicator for whether an incumbent party was mentioned across these three questions without differentiating them. Any remaining 'none' or 'don't know' answers are coded as *not* supporting the incumbent party.

I measure income changes at the household level for the main specifications. Theoretically, the household is the unit at which consumption decisions are made and material wellbeing is experienced, making it the natural "pocketbook" for retrospective evaluations. Technically, many UK tax-benefit instruments are assessed at the benefit-unit or household level, and UKMOD assigns entitlements to the benefit-unit head, creating artificial zero-policy-effects for partners when income is measured individually. Aggregating to the household eliminates this asymmetry while preserving the additive decomposition.

Estimation and identification To examine whether pocketbook voting is policy-based or outcome-based, I regress incumbent vote intention on both total and policy-induced income changes simultaneously (see Appendix E for model specifications). This is equivalent to including policy-induced and non-policy components separately (Wooldridge 2016, p. 126), but is more easily interpretable (I report results for these alternative specifications in Appendix F). Under Kramer's perfect attribution, the coefficient on policy-induced changes should be positive and the coefficient on total changes zero; under Fiorina's indiscriminate pocketbook voting, the reverse. In reality, attribution may be somewhere in between, reflected by a positive coefficient on both.

While the microsimulation model captures the causal effects of policy on disposable incomes by construction, it does not guarantee causal identification of voters' responses to these changes. I follow Tilley, Neundorf and Hobolt (2018) in using lagged dependent variable (LDV) models that explicitly condition on prior partisanship, alongside wave fixed effects and controls. This both accounts for the fact that party support exhibits strong state dependence and for time-varying confounding from, e.g., endogenous policy targeting. Lagged party support dummies are interacted with government period indicators to account for changing incumbent identity (see Appendix E for details). Models further include controls for lagged employment status, gender, age, lagged disposable income, and lagged pocketbook evaluations. I also estimate two-way fixed effects (TWFE) models as a robustness check because LDV and TWFE estimators make different identifying assumptions that typically bracket the causal parameter (Ding and Li 2019). Additional robustness checks in Appendix F vary controls, sample restrictions, and income measurement level.

To account for the fact that policy-eligibility is partly determined at the household level, I cluster standard errors at the level of households (Abadie et al. 2023).[11]

Results

Before presenting model estimates, I descriptively characterize the composition of disposable income changes in terms of the policy-induced and non-policy components. This in turn matters for the degree of attribution needed for pocketbook voting to be policy-sensitive.

Shares of policy-induced and non-policy changes in voters' pocketbooks

The first question of interest is descriptive: how much pocketbook variation is generally attributable to policy changes? To answer this, I first consider overall variation in the two decomposed income change variables, namely respondent-year disposable income changes that are policy-induced or not. Figure 3 shows these distributions, truncated at +/- 600 £/month. Unsurprisingly, there is far more non-policy variation ($SD = 2335$ £/month) than variation from policy changes ($SD = 801$ £/month). When ordered by absolute magnitude, i.e. ignoring the sign of the change, the median non-policy pocketbook change is 185 £/month while the median policy-induced change is just 10 £/month.

Just how large are the components relative to each other? Recall the simulations in Figure 1 showing the importance of this question for policy-sensitivity: the larger the magnitude of non-policy relative to policy-induced income variation, the more correct attributions voters would need to make concordant incumbent sanctions. Figure 4 shows the relative share of policy-induced income changes across respondent-year observations, i.e., the magnitude of policy-induced variation as a share of total variation.[12] As shown, the average share of policy-induced income change is 19% (median = 7%), putting it in the lower end of the scale in Figure 1 (and close to Kramer's

[11] UKHLS only provides unique household identifiers within and not across waves. I therefore construct longitudinal household identifiers based on respondents' first wave in the panel, if they are already in a household, or the first wave in which they joined an existing household.

[12] I measure this by summing the absolute value of each component and calculating the policy-induced change as a percentage of this total: $\frac{|\Delta y_{it}^{policy}|}{|\Delta y_{it}^{policy}|+|\Delta y_{it}^{nonpolicy}|}$.

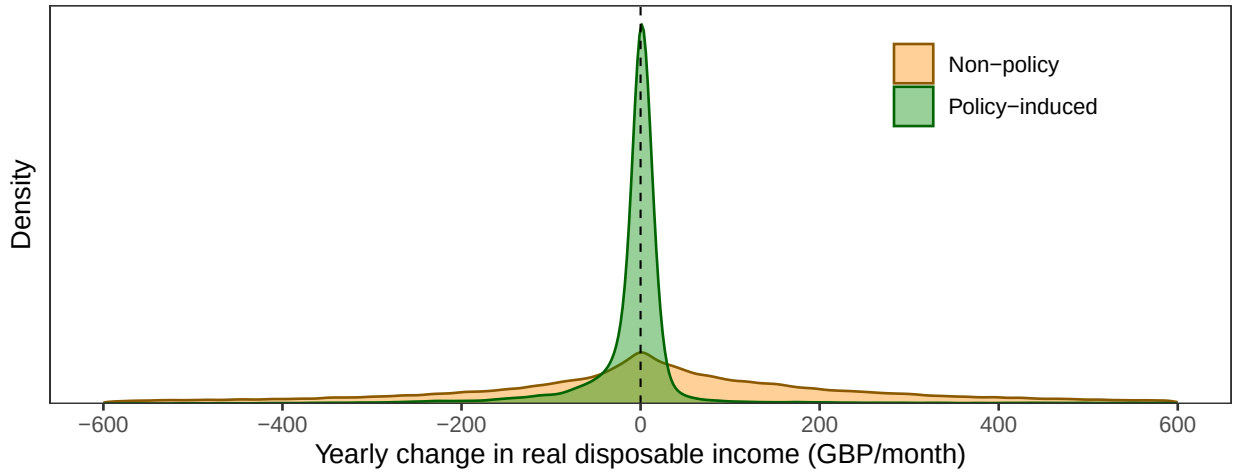


Figure 3: Histogram of the distributions of policy-induced and non-policy disposable income changes at the respondent-year level in the UK from 2010-2019. Truncated at changes of +/- 600 £/month.

informal guess of 0-10%). In just around 12% of observed total income changes is the policy-induced change larger than the non-policy change. As per Figure 1, with this amount of non-policy variation, a relatively high degree of attribution is a necessity for pocketbook voting to result in concordant sanctions.

In sum, citizens' disposable incomes vary substantially from year to year, and for most people, only a modest share of this variation is caused directly by policy changes. Importantly, these results likely represent a high watermark for policy-induced income changes due to the large changes to the UK fiscal system during the decade under study. The key takeaway is that total disposable income changes are not a reliable proxy for the direct income effects of policy. To be policy-sensitive, pocketbook voting must therefore involve voters responding more strongly to policy-induced income changes than other changes. The next section examines whether they do.

How policy-induced pocketbook changes affect incumbent support

I first regress incumbent support on total disposable income changes to validate the basic pocketbook relationship. As shown in the 'total' columns in Table 1, voters who experience positive income changes are indeed more likely to support the incumbent (see in Appendix D for further

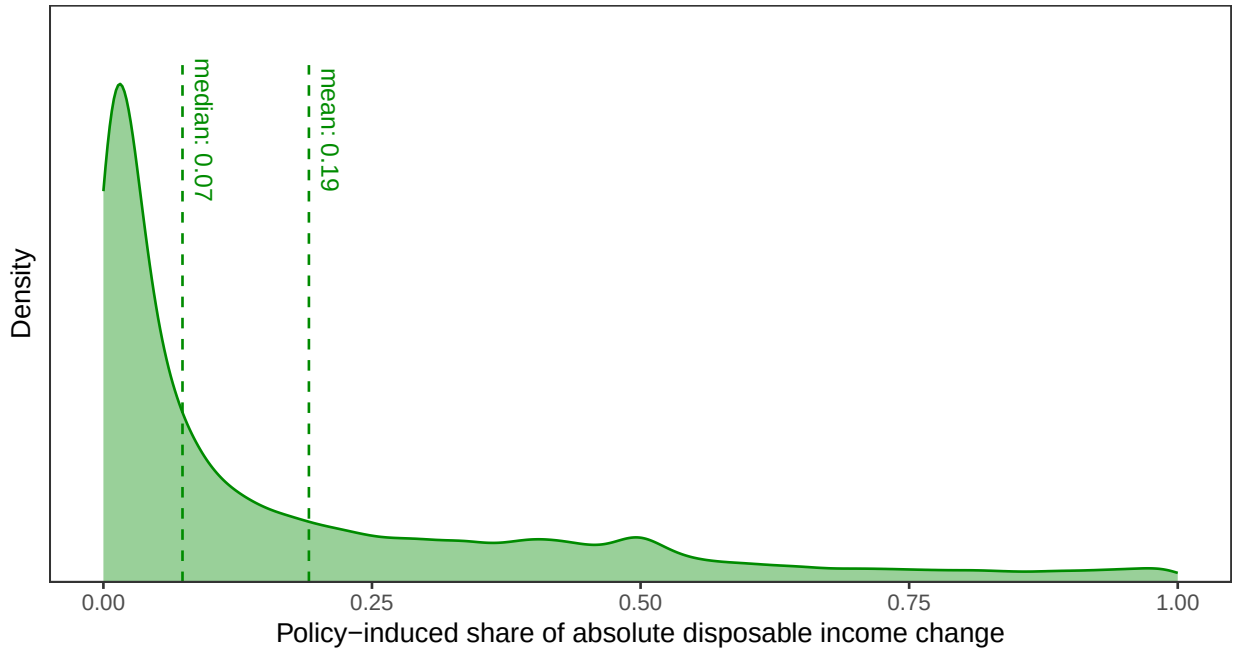


Figure 4: Shares of policy-induced income changes at respondent-year level, as proportion of absolute disposable income change.

validation). The critical question is whether this relationship reflects policy attribution. If voters are more sensitive to policy-induced changes than to other income changes, we should observe a positive coefficient on Δy^{policy} when Δy^{total} is included as a regressor. I test this by regressing incumbent support on both income change variables in the ´decomposed' columns.

Table 1 shows the results of the main regression specifications. As shown, there is little sign of policy attribution in pocketbook voting. The coefficient on total income changes is consistently positive and significant across specifications and it barely changes when policy-induced changes are included separately. The coefficient on policy-induced income changes, on the other hand, is small and not statistically different from zero in any of the main specifications. Because Δy^{total} includes Δy^{policy}, the coefficient on Δy^{policy} in this model captures the additional effect of policy-induced income beyond its contribution to total income. The null coefficient thus implies that there is on average no difference between voters' responsiveness to policy-induced and other income changes.[13]

[13]This differential interpretation is exact for the untransformed specification (columns 1–2), where the additive identity

Table 1: Relationship between policy-induced and total disposable income changes and incumbent support.

	Income changes (£1000/mo)		Income changes (asinh)	
	Total	Decomposed	Total	Decomposed
Δy^{total}	0.15 (0.03)***	0.16 (0.05)**	0.06 (0.02)**	0.06 (0.02)**
Δy^{policy}		−0.03 (0.15)		0.00 (0.03)
N	91 295	91 295	91 295	91 295
Std.Errors	household	household	household	household
Lagged DV	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Note. Regression table of lagged dependent variable models with household-clustered errors and lagged control variables. The outcome is incumbent support (0-100). Independent variables are yearly changes in monthly disposable household income, in real terms (raw £1000/month and asinh transformed). Policy-induced changes reflect net real changes in taxes and social contributions paid and transfers received due to changes in simulated policies. Total income changes reflect the full change in disposable income, encompassing both policy-induced and non-policy components. Full model results including all covariate estimates are reported in Table E1 (Appendix E). * p<.05, ** p<.01, *** p<.001.

The null coefficient does not, however, rule out some degree of differential responsiveness. Just how much policy attribution is consistent with the data? I quantify this as the ratio of voters' sensitivity to policy-induced income changes over their sensitivity to non-policy changes, i.e. the factor by which voters weight the policy component relative to other income variation. A ratio of 1 corresponds to Fiorina's equal-weighting model; higher ratios indicate progressively stronger attribution. The point estimate, recovered from the decomposition specifications in Table 1, is approximately 1.01 — consistent with Fiorina's outcome-based model of equal weighting.[14]

Following Rainey (2014), I use 90% confidence intervals to assess whether the data can rule

$\Delta y^{\text{total}} = \Delta y^{\text{policy}} + \Delta y^{\text{nonpolicy}}$ holds. Under the asinh transformation (columns 3–4), the identity is approximate since $\text{asinh}(a+b) \neq \text{asinh}(a) + \text{asinh}(b)$. In practice, the distortion is minimal: policy-induced changes are small (median $\approx$ £10/month), and for small values asinh is approximately linear, so the additive relationship is largely preserved for the policy component. The null obtains under both transformations regardless.

[14]The ratio and its confidence interval are derived via the delta method from the specifications in Table 1; see Appendix M for details.

out meaningful attribution[15]. The 90% confidence interval for the attribution ratio, [0.24, 1.77], rules out the higher attribution ratios (5:1, 10:1) from Figure 1, though it cannot exclude modest differential weighting up to approximately 2:1. Given that non-policy variation accounts for around 80% of observed income variation, placing us in the upper end of Figure 1, even a ratio of 2:1 would not produce meaningfully policy-sensitive pocketbook voting per the simulations. These results are therefore inconsistent with pocketbook voting being even moderately policy-sensitive.

This core finding of no substantial differential policy effect holds across various alternative specifications, including two-way fixed effects instead of lagged dependent variables, removing controls, including dummies for extreme income changes, using quadratic terms, including lagged income change variables, excluding the highest earners and individuals eligible for very small benefits, and using individual-level rather than household-level income changes (see Appendix F). I further find no significant differences in these relationships by the party in power, although these tests are imperfect (see Appendix I). Additional robustness checks confirm that results are unchanged when using survey weights (Appendix G), restricting the sample to households with stable composition (Appendix H), excluding post-Brexit referendum waves (Appendix I), and using cumulative multi-year income changes instead of year-on-year changes (Appendix K). Both the Brexit exclusion and governing-period analyses are reported together in Appendix I.

I further test whether voters engage in cruder forms of attribution based on the *source* of income changes – responding more to changes in taxes paid or transfers received than to market income changes, regardless of whether these reflect policy reforms or changes in personal circumstances. As reported in Appendix O, there is little evidence of even this more pragmatic form of attribution: voters appear to treat a pay rise and a tax cut equivalently when evaluating incumbents, with no systematically stronger response to tax or transfer changes.

[15]The 90% confidence intervals on the policy-induced coefficients in Table 1 are [-0.28; 0.21] for the raw income changes and [-0.05; 0.05] for the asinh income changes. See Table L1 (Appendix L) for full results.

Heterogeneous effects

Even if voters on average respond equally to all income changes, incumbents could still face policy-sensitive sanctioning from specific constituencies. For instance, politically attentive voters could successfully distinguish policy-induced income changes while others do not. Such heterogeneity could produce the aggregate null documented above while still shaping incumbent incentives. It is also possible that the null reflects opposing subgroup effects canceling out. To test this, I estimate heterogeneous effects across six moderators: age, political interest, lagged party support, respondent-level average disposable income, economic status, and the share of total income change attributable to policy (policy-induced share). Figure 5 summarizes the results. For continuous moderators, marginal effects from linear interaction models are plotted across the moderator's observed range; for categorical moderators, separate regressions are estimated for each group. Full interaction estimates are reported in Table J1 and Table J2 (Appendix J).

The lack of attribution generalizes across all subgroups examined. The main result reflects a general absence of attribution rather than mixed responsiveness canceling out across the electorate. Notably, even highly politically interested voters – who are most likely to follow policy developments and media coverage of fiscal changes – show no greater sensitivity to policy-induced income changes. This is consistent with attribution being fundamentally challenging given the structure of income variation, rather than a matter of political sophistication or attentiveness.

The policy-induced share (the proportion of total income change attributable to policy) varies considerably across voters and provides a more direct window into the attribution mechanism. A signal-extraction logic would predict that voters find it easier to identify policy-induced changes when these are not swamped by other income fluctuations. Consistent with this, the interaction between policy-induced income changes and the policy-induced share is positive and statistically significant (Appendix J): as Figure 5 shows, responsiveness to policy-induced income changes increases with the policy-induced share. This pattern is also consistent with a nonlinear salience effect, since the policy-induced share is mechanically correlated with the absolute magnitude of the policy change; the interaction cannot fully distinguish signal extraction from a threshold re-

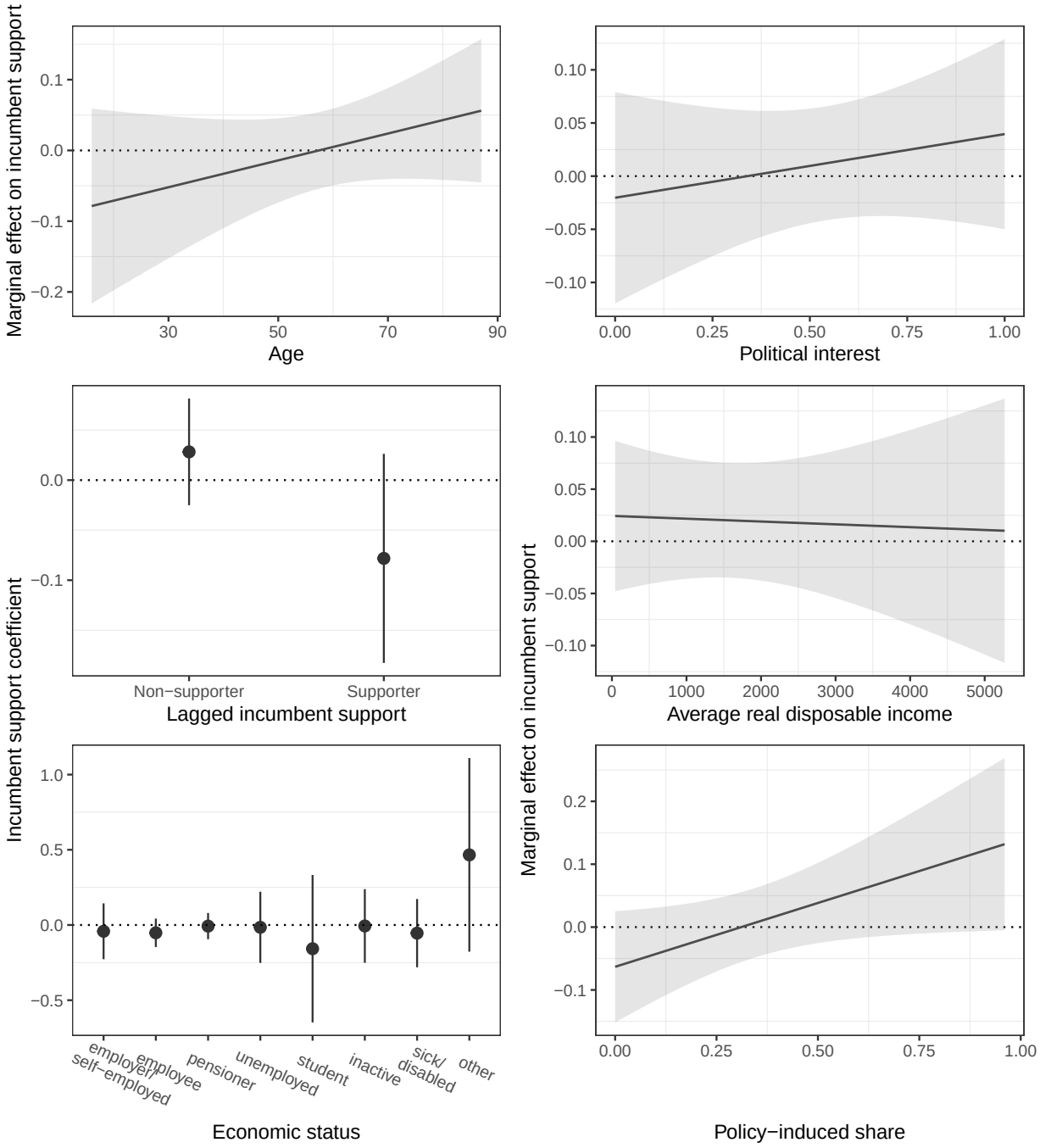


Figure 5: Heterogeneous effects of decomposed income changes on incumbent support. Coefficients and marginal effects for Δy^{policy} (policy-induced income changes). For continuous moderators, lines show marginal effects across the moderator's observed range; for categorical moderators, points show group-specific estimates from separate regressions. 'Policy-induced share' is the share of the total income change attributable to policy.

sponse to large policy shocks. Yet even at high values of the policy-induced share, the estimated responsiveness remains modest – and for the vast majority of voters, whose policy-induced share is low, policy-induced income changes are effectively invisible against the backdrop of other income variation. Whether this pattern reflects a perceptual threshold – voters simply cannot detect small policy changes against background income variation – or an attribution failure – voters detect the changes but cannot identify their source – cannot be fully distinguished with this design, though both imply weak accountability in practice.

Discussion and Conclusion

Pocketbook voting is widely seen as a mechanism for holding incumbents accountable: voters reward governments when their finances improve and punish them when they decline (Fiorina 1978; Key 1966; Kramer 1983). But for this mechanism to discipline incumbent policy, voters must respond reliably to the policy-induced component of their income changes – what I term policy-sensitivity. Using microsimulation models to decompose income changes into policy-induced and non-policy components, this paper provides a direct test of whether voters make this distinction, moving beyond single-case studies to estimate how reliably voters sanction incumbents for the net income effects of policy on average.

The results offer little support for policy-sensitive pocketbook voting. For most voters, policy-induced income variation is small relative to other sources of income change, accounting for roughly a fifth of total variation on average. Voters do not compensate by placing greater weight on the policy-induced component: the data cannot reject equal weighting of policy-induced and non-policy income changes, and rule out differential responsiveness above roughly 2:1. Mapped onto the theoretical framework in Figure 1, these estimates place the UK electorate in the region where outcome-based pocketbook voting rarely produces concordant sanctions more often than chance. This lack of attribution holds consistently across demographic subgroups and levels of political interest, and is consistent with a signal-extraction logic: voters respond more to policy-induced income changes only when these are not swamped by other income fluctuations – a condition that

holds for few voters in practice.

Some limitations to the analysis remain. First, I focus on policy changes that have direct effects on disposable incomes. Yet, electoral accountability, broadly speaking, is about more than *direct* income effects, with e.g. the indirect effects of trade policy, industrial policy, and public goods provision notably being outside the scope of my analysis. This is partially driven by the fact that the pocketbook effects of such policies is fraught with uncertainty. More importantly, there is little reason to expect voters to be more responsive in the domain of indirect income effects. As such, my focus on direct income effects is a more likely case to observe attribution. Still, I cannot firmly conclude that my results generalize to other policy domains. Similarly, the exclusion of State Pension and disability benefits – not simulated from rules in UKMOD – means the analysis does not capture certain transfer instruments where policy feedback may be stronger.

Second, I exploit a unique opportunity to link survey measures of incumbent support and microsimulated income changes in present-day UK. In many ways, this happens to be a good case to study policy-sensitivity of pocketbook voting given the dramatic transformations of the UK's fiscal system at the time, generating lots of useful variation in the independent variables. Still, it naturally raises the question of how my findings generalize to other contexts. I provide evidence that my findings are fairly robust across subgroups in the electorate, suggesting that the lack of attribution is pervasive and not a feature of the composition of the British electorate. I further find no indication that attribution was any different under the Labour government. Still, I mostly study a period of welfare state retrenchment under successive center-right governments, which may be special in other ways. That said, given the extent to which the government's austerity' program was politicized, this is unlikely to be a case of unusually *low* public responsiveness to policy effects. It is important to note, however, that my results are not inconsistent with negative electoral responses to austerity policies as such. The UK Conservatives' austerity policies encompassed cuts to various public services and through fiscal multipliers in local economies may have affected voting behavior via their effects on social groups or communities rather than individual pocketbooks (Fetzer 2019; Gaikwad, Genovese and Tingley 2022). More research should examine such

group-level retrospective voting behavior.

Third, this paper has focused on estimating the overall relationship between income policy and pocketbook behavior and only secondarily on psychological mechanisms. To understand exactly how and when voters attribute income changes differently, narrower case studies and experiments are typically more useful. Existing work already provides important insights on this, and my results are not inconsistent with any of these psychological theories *per se*. Future studies should examine these questions further. Nonetheless, if I had a richer survey dataset than the UKHLS available for linking with UKMOD, it would have allowed me to get more definitive evidence for the psychological mechanisms at play.

These caveats notwithstanding, my results have several important implications. First and foremost, they extend and cast new light on existing empirical work on pocketbook attribution. Several studies suggest that citizens seem more responsive to financial changes that they subjectively attribute to government action (Abramowitz, Lanoue and Ramesh 1988; Feldman 1982; Lau and Sears 1981; Marsh and Tilley 2010; Tilley, Neundorf and Hobolt 2018). Likewise, research on policy feedback has in some cases found voters to respond to economic policies that affect them (Kogan 2021; Manacorda, Miguel and Vigorito 2011; Pop-Eleches 2012; Rendleman and Yoder 2024; Zucco Jr. 2013). Despite this, I find that voters do not *on average* sanction incumbents for policy-induced income changes. This suggests that although voters evidently aspire to link income changes to policy, and sometimes do so, the average pattern is one of indiscriminate responsiveness. When voters' sanctions happen to align with policy-induced income changes, it appears to be because their "rough justice" (Fiorina 1981, p.4) delivered concordant sanctions in cases where total income changes tracked the policy component rather than because voters distinguished the two. These instances of policy feedback, while important for understanding specific policies, thus mask a broader pattern of lacking attribution.

Second, the study demonstrates the value of linking microsimulated disposable incomes to individual-level attitudes. Rarely used in political science (Avram and Popova 2022; Elkjær and Mushövel 2023), these models offer a uniquely detailed account of how policy changes shape

voters' financial situations while covering wholesale changes to the fiscal system. By leveraging them to estimate the net impact of an incumbent's full policy record, this paper provides a novel approach to studying pocketbook voting and policy feedback. Future research can build on this approach to analyze electoral reactions to specific policy changes, shedding further light on how voters respond to specific types of pocketbook changes.

Finally, by estimating the extent of policy-sensitivity in pocketbook voting, this study speaks to broader debates on electoral politics and democratic accountability. One key implication concerns changes in party coalitions. If voters do not differentially respond to the policy-induced component of their income changes, parties may struggle to earn electoral credit simply by enacting material benefits. This casts doubt on the effectiveness of strategies like the US Democratic party's so-called 'deliverism' approach in the 2024 Presidential Election, which sought to win voters organically by delivering tangible material gains (Bhargava, Shams and Hanbury 2024). My findings suggest that its apparent failure is less an anomaly than a reflection of a broader pattern: in a world of low policy-sensitivity, material benefits do not automatically translate into reliable political rewards. This has important implications for how parties approach governance and campaigning. Rather than assuming that good policy outcomes will speak for themselves, parties may need to invest heavily in communication strategies that make policy effects visible and attributable. The Schumer lament that opens this paper – that legislators assumed voters would notice their work – finds empirical grounding here: when policy-induced income changes are small relative to other variation, policy effects do not speak for themselves. Alternatively, parties might focus on highly salient, easily attributable policies even if these are not the most economically efficient approaches.

Low policy-sensitivity also reshapes expectations about strategic incumbent behavior. In models where pocketbook voting is policy-sensitive, incumbents face strong incentives for tactical redistribution to electorally pivotal groups rather than policies that foster broad-based growth (Battaglini and Coate 2007, 2008; Maskin and Tirole 2019). If voters are largely indifferent to the policy-induced component of income changes, such strategies are unlikely to generate reliable electoral returns. This does not eliminate all electoral incentives for redistributive policies,

but it fundamentally alters them. The signal-extraction finding illustrates how: voters do respond more to policy-induced changes when these are large relative to other income fluctuations. Incumbents seeking electoral returns from redistributive policy may therefore find the highest payoff in targeting voters for whom policy changes represent a large share of total income variation — though which voters meet this condition is not straightforwardly predicted by income level alone, as lower-income voters may also face greater non-policy income volatility. Another possibility is that voters respond more to community- or group-level economic conditions than to personal income changes, which could generate similar tactical incentives (Dausgaard 2025; Drazen and Eslava 2006). Future theoretical work should derive the incumbent incentives that arise under the conditions documented here: policy-insensitive pocketbook voting with modest policy-induced income variation.

These results do not eliminate the possibility of electoral manipulation under specific conditions. Research on policy feedback suggests that highly visible, easily traceable policies can shape voter behavior (Hamel 2024; Shanks-Booth and Mettler 2019), and the signal-extraction finding implies that incumbents retain some control over which policies voters notice. But in ordinary times, when policy-induced income changes are a small fraction of voters' overall income variation, pocketbook voting operates largely as Fiorina envisioned: voters exert "rough justice" based on total income changes, and their sanctions align with the effects of policy changes mostly incidentally. The disciplining power of pocketbook voting may therefore be considerably weaker than standard models of electoral accountability assume.

References

Abadie, Alberto, Susan Athey, Guido W Imbens and Jeffrey M Wooldridge. 2023. "When should you adjust standard errors for clustering?" *The Quarterly Journal of Economics* 138(1):1–35.

Abeler, Johannes and Simon Jäger. 2015. "Complex tax incentives." *American Economic Journal: Economic Policy* 7(3):1–28.

Abramowitz, Alan I., David J. Lanoue and Subha Ramesh. 1988. "Economic Conditions, Causal Attributions, and Political Evaluations in the 1984 Presidential Election." *The Journal of Politics* 50(4):848–863.

Achen, Christopher H and Larry M Bartels. 2016. *Democracy for Realists*. Princeton University Press.

Ashworth, Scott and Ethan Bueno de Mesquita. 2014. "Is Voter Competence Good for Voters?: Information, Rationality, and Democratic Performance." *American Political Science Review* 108(3):565–587.

Ashworth, Scott, Ethan Bueno de Mesquita and Amanda Friedenberg. 2018. "Learning about Voter Rationality." *American Journal of Political Science* 62(1):37–54.

Avram, Silvia and Daria Popova. 2022. "Do taxes and transfers reduce gender income inequality? Evidence from eight European welfare states." *Social Science Research* 102:102644.

Bargain, Olivier and Tim Callan. 2010. "Analysing the Effects of Tax-Benefit Reforms on Income Distribution: A Decomposition Approach." *The Journal of Economic Inequality* 8(1):1–21.

Battaglini, Marco and Stephen Coate. 2007. "Inefficiency in Legislative Policymaking: A Dynamic Analysis." *American Economic Review* 97(1):118–149.

Battaglini, Marco and Stephen Coate. 2008. "A Dynamic Theory of Public Spending, Taxation, and Debt." *American Economic Review* 98(1):201–236.

Bechtel, Michael M. and Roman Liesch. 2020. "Reforms and Redistribution: Disentangling the Egoistic and Sociotropic Origins of Voter Preferences." *Public Opinion Quarterly* 84(1):1–23.

Bellemare, Marc F. and Casey J. Wichman. 2020. "Elasticities and the Inverse Hyperbolic Sine Transformation." *Oxford Bulletin of Economics and Statistics* 82(1):50–61.

Bhargava, Deepak, Shahrzad Shams and Harry Hanbury. 2024. "The Death of Deliverism: Why Policy Alone Is Not Enough." 33(1):6–16.
URL: *https://doi.org/10.1177/10957960231221761*

Bhargava, Saurabh and Dayanand Manoli. 2015. "Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment." *American Economic Review* 105(11):3489–3529.

Bourquin, Pascale, Agnes Norris Keiller and Tom Waters. 2019. The Distributional Impact of Personal Tax and Benefit Reforms, 2010 to 2019. IFS Briefing Note BN270 Institute for Fiscal Studies.
URL: *https://ifs.org.uk/publications/distributional-impact-personal-tax-an d-benefit-reforms-2010-2019*

Brody, Richard A. and Paul M. Sniderman. 1977. "From Life Space to Polling Place: The Relevance of Personal Concerns for Voting Behavior." *British Journal of Political Science* 7(3):337–360.

Bronka, Patryk, Daria Popova and Matteo Richiardi. 2023. UK Household Longitudinal Study (UKHLS) – Understanding Society. Technical Report 7/23 CEMPA.

Ciobanu, Costin. 2024. "The electoral risks of austerity." *European Journal of Political Research* 63(1):348–369.

Conover, Pamela Johnston. 1984. "The Influence of Group Identifications on Political Perception and Evaluation." *The Journal of Politics* 46(3):760–785.

Dausgaard, Christoffer Hentzer. 2025. "You and Whose Economy? Group-Based Retrospection in Economic Voting." *American Journal of Political Science* .

Ding, Peng and Fan Li. 2019. "A bracketing relationship between difference-in-differences and lagged-dependent-variable adjustment." *Political Analysis* 27(4):605–615.

Drazen, Allan and Marcela Eslava. 2006. Pork Barrel Cycles. Technical Report w12190 National Bureau of Economic Research Cambridge, MA: .

Duch, Raymond M. and Randolph T. Stevenson. 2008. *The Economic Vote: How Political and Economic Institutions Condition Election Results*. Cambridge University Press.

Elkjær, Mads Andreas and Fabian Mushövel. 2023. "Democratic Compensation or Winner-Take-All Domination? The Distributional Effects of Tax-and-Transfer Policies in 21st Century Europe." Working paper.
URL: *https://madselkjaer.com/wp-content/uploads/2023/11/elkjaer_mushovel_2023.pdf*

Equality and Human Rights Commission. 2018. The Cumulative Impact of Tax and Welfare Reforms. Technical report Equality and Human Rights Commission. Research by Jonathan Portes and Howard Reed.
URL: *https://www.equalityhumanrights.com/sites/default/files/cumulative-impact-assessment-report.pdf*

Feldman, Stanley. 1982. "Economic Self-Interest and Political Behavior." *American Journal of Political Science* 26(3):446–466.

Feldman, Stanley. 1984. "Economic Self-Interest and the Vote: Evidence and Meaning." *Political Behavior* 6(3):229–251.

Ferejohn, John. 1986. "Incumbent Performance and Electoral Control." *Public Choice* 50(1/3):5–25.

Fetzer, Thiemo. 2019. "Did Austerity Cause Brexit?" 109(11):3849–3886.
URL: *https://pubs.aeaweb.org/doi/10.1257/aer.20181164*

Fiorina, Morris P. 1978. "Economic Retrospective Voting in American National Elections: A Micro-Analysis." *American Journal of Political Science* 22(2):426–443.

Fiorina, Morris P. 1981. *Retrospective Voting in American National Elections*. Yale University Press.

Fowler, Anthony and Andrew B. Hall. 2018. "Do Shark Attacks Influence Presidential Elections? Reassessing a Prominent Finding on Voter Competence." *The Journal of Politics* 80(4):1423–1437.

Fowler, Anthony and B. Pablo Montagnes. 2015. "College Football, Elections, and False-Positive Results in Observational Research." *Proceedings of the National Academy of Sciences* 112(45):13800–13804.

Fowler, Anthony and B. Pablo Montagnes. 2023. "Distinguishing between False Positives and Genuine Results: The Case of Irrelevant Events and Elections." *The Journal of Politics* 85(1):304–309.

Gaikwad, Nikhar, Federica Genovese and Dustin Tingley. 2022. "Creating Climate Coalitions: Mass Preferences for Compensating Vulnerability in the World's Two Largest Democracies." *American Political Science Review* 116(4):1165–1183.

Gomez, Brad T. and J. Matthew Wilson. 2001. "Political Sophistication and Economic Voting in the American Electorate: A Theory of Heterogeneous Attribution." *American Journal of Political Science* 45(4):899–914.

Graham, Matthew H., Gregory A. Huber, Neil Malhotra and Cecilia Hyunjung Mo. 2023. "Irrelevant Events and Voting Behavior: Replications Using Principles from Open Science." *The Journal of Politics* 85(1):296–303.

Hacker, Jacob S. 2004. "Privatizing risk without privatizing the welfare state: The hidden politics of social policy retrenchment in the United States." *American Political Science Review* 98(2):243–260.

Hamel, Brian T. 2024. "Traceability and Mass Policy Feedback Effects." *American Political Science Review* pp. 1–16.

Hasell, Joe. 2023. "Income inequality before and after taxes: how much do countries redistribute income?" *Our World in Data* . https://ourworldindata.org/income-inequality-before-and-after-taxes.

Healy, Andrew J., Mikael Persson and Erik Snowberg. 2017. "Digging into the Pocketbook: Evidence on Economic Voting from Income Registry Data Matched to a Voter Survey." *American Political Science Review* 111(4):771–785.

Healy, Andrew J. and Neil Malhotra. 2010. "Random Events, Economic Losses, and Retrospective Voting: Implications for Democratic Competence." *Quarterly Journal of Political Science* 5(2):193–208.

Healy, Andrew J., Neil Malhotra and Cecilia Hyunjung Mo. 2010. "Irrelevant Events Affect Voters' Evaluations of Government Performance." *Proceedings of the National Academy of Sciences* 107(29):12804–12809.

HM Treasury. 2010. Budget 2010. Technical Report HC 61 HM Treasury.
URL: *https://assets.publishing.service.gov.uk/media/5a7b7f37ed915d1a79023a82/0061.pdf*

HM Treasury. 2015. "Summer Budget 2015.". Accessed August 26, 2025.
URL: *https://www.gov.uk/government/publications/summer-budget-2015/summer-budget-2015*

Hobolt, Sara, James Tilley and Susan Banducci. 2013. "Clarity of responsibility: How government cohesion conditions performance voting." *European journal of political research* 52(2):164–187.

Holmström, Bengt. 1979. "Moral Hazard and Observability." *The Bell Journal of Economics* 10(1):74–91.
URL: *http://www.jstor.org/stable/3003320*

House of Commons Library. 2015. Summer Budget 2015: A summary. Technical report House of Commons Library. Accessed August 26, 2025.
URL: *https://commonslibrary.parliament.uk/research-briefings/cbp-7251/*

House of Commons Library. 2016. Universal Credit Changes from April 2016. Briefing Paper CBP-7446 House of Commons Library.
URL: *https://commonslibrary.parliament.uk/research-briefings/cbp-7446/*

Huber, Gregory A., Seth J. Hill and Gabriel S. Lenz. 2012. "Sources of Bias in Retrospective Decision Making: Experimental Evidence on Voters' Limitations in Controlling Incumbents." *American Political Science Review* 106(4):720–741.

Key, V. O. 1966. *The Responsible Electorate: Rationality in Presidential Voting, 1936–1960*. Harvard University Press.

Kiewiet, D. Roderick and Douglas Rivers. 1984. "A Retrospective on Retrospective Voting." *Political Behavior* 6(4):369–393.

Kogan, Vladimir. 2021. "Do Welfare Benefits Pay Electoral Dividends? Evidence from the National Food Stamp Program Rollout." *The Journal of Politics* 83(1):58–70.

Kramer, Gerald H. 1983. "The Ecological Fallacy Revisited: Aggregate- versus Individual-level Findings on Economics and Elections, and Sociotropic Voting." *The American Political Science Review* 77(1):92–111.

Larsen, Martin Vinæs. 2019. "Is the relationship between political responsibility and electoral accountability causal, adaptive and policy-specific?" *Political Behavior* 41(4):1071–1098.

Larsen, Martin Vinæs. 2021. "How Do Voters Hold Politicians Accountable for Personal Welfare? Evidence of a Self-Serving Bias." *The Journal of Politics* 83(2):740–752.

Lau, Richard R. and David O. Sears. 1981. "Cognitive Links between Economic Grievances and Political Responses." *Political Behavior* 3(4):279–302.

Lupia, Arthur and Mathew D McCubbins. 1998. *The democratic dilemma: Can citizens learn what they need to know?* Cambridge University Press.

Manacorda, Marco, Edward Miguel and Andrea Vigorito. 2011. "Government Transfers and Political Support." *American Economic Journal: Applied Economics* 3(3):1–28.

Marsh, Michael and James Tilley. 2010. "The Attribution of Credit and Blame to Governments and Its Impact on Vote Choice." *British Journal of Political Science* 40(1):115–134.

Maskin, Eric and Jean Tirole. 2019. "Pandering and Pork-Barrel Politics." *Journal of Public Economics* 176:79–93.

McKenzie, David. 2023. "Interpreting Treatment Effects on an Inverse Hyperbolic Sine Outcome Variable and Alternatives.".
URL: *https://blogs.worldbank.org/en/impactevaluations/interpreting-treatment-effects-inverse-hyperbolic-sine-outcome-variable-and*

Mettler, Suzanne. 2011. *The submerged state: How invisible government policies undermine American democracy*. University of chicago Press.

Miller, Michael K. 2013. "For the Win! The Effect of Professional Sports Records on Mayoral Elections." *Social Science Quarterly* 94(1):59–78.

Mutz, Diana C. 1994. "Contextualizing Personal Experience: The Role of Mass Media." *The Journal of Politics* 56(3):689–714.

National Audit Office. 2013. Council Tax Support. Technical Report HC 882, Session 2013–14 National Audit Office.
URL: *https://www.nao.org.uk/wp-content/uploads/2013/12/10316-001-Council-Tax-Book.pdf*

Office for Budget Responsibility. 2019. Welfare Trends Report – December 2019. Technical report Office for Budget Responsibility.
URL: *https://obr.uk/docs/dlm_uploads/Welfare_trends_report_December_2019.pdf*

Osborne, George. 2015. "Chancellor George Osborne's Summer Budget 2015 speech.". Accessed August 26, 2025.
URL: *https://www.gov.uk/government/speeches/chancellor-george-osbornes-summer-budget-2015-speech*

Paulus, Alari and Iva Valentinova Tasseva. 2020. "Europe Through the Crisis: Discretionary Policy Changes and Automatic Stabilizers." *Oxford Bulletin of Economics and Statistics* 82(4):864–888.

Peters, B Guy. 1991. "The politics of taxation." *A Comparative Perspective, Cambridge* pp. 58–64.

Pop-Eleches, Cristian. 2012. "Targeted Government Spending and Political Preferences." *Quarterly Journal of Political Science* 7(3):285–320.

Powell, G. Bingham and Guy D. Whitten. 1993. "A Cross-National Analysis of Economic Voting: Taking Account of the Political Context." *American Journal of Political Science* 37(2):391–414.

Rainey, Carlisle. 2014. "Arguing for a Negligible Effect." *American Journal of Political Science* 58(4):1083–1091.

Reis, Sara and Iva Tasseva. 2020. EUROMOD/UKMOD Country Report 2018-2024. Technical report UKMOD.

Rendleman, Hunter and Jesse Yoder. 2024. "Do Government Benefits Affect Officeholders' Electoral Fortunes? Evidence from State Earned Income Tax Credits." *American Political Science Review* p. 1–18.

Richiardi, Matteo, Diego Collado and Daria Popova. 2021. "UKMOD – A New Tax- Benefit Model for the Four Nations of the UK." *International Journal of Microsimulation* 14(1):1–10.

Sances, Michael W. 2017. "Attribution Errors in Federalist Systems: When Voters Punish the President for Local Tax Increases." *The Journal of Politics* 79(4):1286–1301.

Shanks-Booth, Delphia and Suzanne Mettler. 2019. "The Paradox of the Earned Income Tax Credit: Appreciating Benefits but Not Their Source." *Policy Studies Journal* 47(2):300–323.

Soss, Joe and Sanford F. Schram. 2007. "A Public Transformed? Welfare Reform as Policy Feedback." 101(1):111–127.

Thurley, Djuna and Richard Keen. 2023. State Pension Triple Lock. Research Briefing CBP-7812 House of Commons Library.
URL: *https://commonslibrary.parliament.uk/research-briefings/cbp-7812/*

Tilley, James, Anja Neundorf and Sara B. Hobolt. 2018. "When the Pound in People's Pocket Matters: How Changes to Personal Financial Circumstances Affect Party Choice." *The Journal of Politics* 80(2):555–569.

UK Parliament. 2012. "Local Government Finance Act 2012: Explanatory Notes.".
URL: *https://www.legislation.gov.uk/ukpga/2012/17/notes/division/3/2*

Wolfers, Justin. 2002. "Are Voters Rational? Evidence from Gubernatorial Elections.".

Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. Cambridge, MA: MIT Press.

Wooldridge, Jeffrey M. 2016. *Introductory Econometrics: A Modern Approach 6rd ed.* Cengage learning.

Zucco Jr., Cesar. 2013. "When Payouts Pay Off: Conditional Cash Transfers and Voting Behavior in Brazil 2002–10." *American Journal of Political Science* 57(4):810–822.

Appendix

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A Simulations of policy-sensitivity

Figure 1 is generated by simulating the policy-induced and non-policy income variation as two random variables that I assume to be i) normally distributed, ii) zero-centered, and iii) uncorrelated:

$$\begin{aligned} \text{policy} &\sim \mathcal{N}(0, \delta_s) \\ \text{nonpolicy} &\sim \mathcal{N}(0, \delta_n) \end{aligned} \tag{2}$$

The magnitude of non-policy income variation is then defined as:

$$\text{non-policy magnitude} = \frac{\delta_n}{\delta_s + \delta_n} \tag{3}$$

where δ_s denotes the standard deviation. I then simulate electorates for values of non-policy magnitude between 1 and 100, which goes on the x-axis. To obtain the curves, I specify five decision rules that determine how voters cast a binary vote either for or against the incumbent. At either extreme is Kramer and Fiorina, with the other three rules falling somewhere in between. They are all specified using Equation 1 by varying the β weights as follows:

$$\begin{aligned} \text{Kramer } (\beta_s > 0, \beta_n = 0) \quad & \begin{cases} 1, & \text{if policy} > 0 \\ 0, & \text{otherwise} \end{cases} \\ \text{Fiorina } (\beta_s = \beta_n > 0) \quad & \begin{cases} 1, & \text{if policy} + \text{nonpolicy} > 0 \\ 0, & \text{otherwise} \end{cases} \\ \beta_s = 2 \times \beta_n > 0 \quad & \begin{cases} 1, & \text{if } 2 \times \text{policy} + \text{nonpolicy} > 0 \\ 0, & \text{otherwise} \end{cases} \\ \beta_s = 5 \times \beta_n > 0 \quad & \begin{cases} 1, & \text{if } 5 \times \text{policy} + \text{nonpolicy} > 0 \\ 0, & \text{otherwise} \end{cases} \\ \beta_s = 10 \times \beta_n > 0 \quad & \begin{cases} 1, & \text{if } 10 \times \text{policy} + \text{nonpolicy} > 0 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

I then calculate, under each of these rules, the share of voters who make incumbent sanctions that align with policy-effects, i.e. voters whose incumbent vote decision is consistent with the sign on their policy-induced component.

B Timing of fiscal changes and survey waves

The budget of the United Kingdom is an annual budget set by the treasury for the following financial year. Budgets are usually set once a year and last for the duration of the financial year running from 1 April to 31 March. Budgets have historically been announced in March less than a month before the start of the new financial year. This practice changed in 2017 where budgets were released 4-5 months earlier, in October or November. Thus, for the 2010-2019 period under study, budgets were announced shortly before implemented policy changes in 2010-2017, and announced in the preceding Autumn for the 2018 and 2019 policy changes.

One exception to the pattern was in 2015, where an additional budget was announced in July, four months after the March budget, due to the change of government. Due to the odd timing of this July budget, some measures took immediate effect on the day of announcement, including changes to carried interest taxation for fund managers and modifications to Controlled Foreign Company rules (HM Treasury 2015; Osborne 2015). However, many of the headline policies were scheduled for implementation at the start of the following financial year in April 2016, including the introduction of the National Living Wage at £7.20 for workers aged 25 and above, the replacement of the Dividend Tax Credit with a £5,000 tax-free Dividend Allowance, and the increase in the Employment Allowance from £2,000 to £3,000 (HM Treasury 2015; House of Commons Library 2015). Other significant measures had even longer implementation timescales, such as the planned reductions in corporation tax rates in 2017 and the phased introduction of restrictions on landlord tax relief starting in 2017 (HM Treasury 2015).

Figure B1 shows how survey respondents in the UKHLS are distributed across waves.

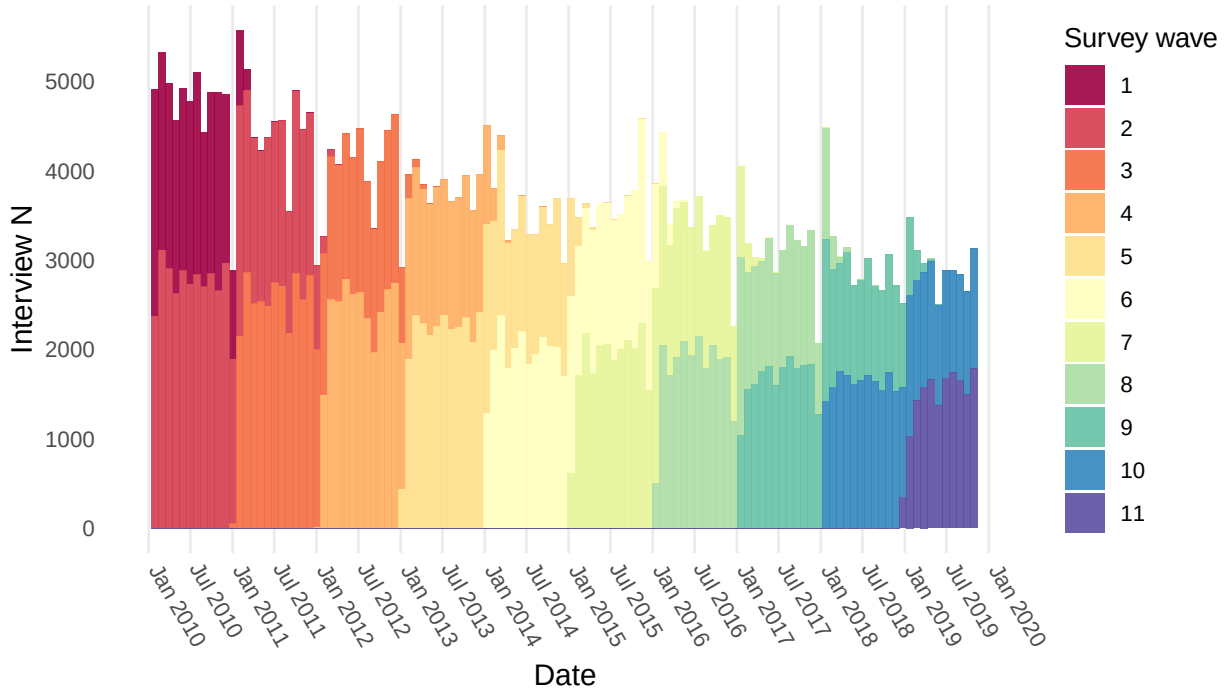


Figure B1: How survey respondents are distributed across waves of the UKHLS.

C UK Tax-and-Transfer Policy Changes, 2010–2019

Table C1 provides a fiscal-year timeline of major UK tax-and-transfer policy changes from 2010–11 through 2019–20, indicating which changes are included in the paper's simulated treatment variable.

As indicated in the rightmost column of Table C1, the simulated treatment variable captures policy changes that produce heterogeneous individual-level effects – primarily income tax parameters, National Insurance contributions (employee and self-employed), means-tested benefits (including Universal Credit, Tax Credits, and Housing Benefit), and Child Benefit rules. The scope and distributional profile of these reforms are documented in detail by Bourquin, Norris Keiller and Waters (2019) and Equality and Human Rights Commission (2018).

The treatment variable excludes two categories of policy changes. First, certain benefits – notably State Pension and disability benefits – are present in UKMOD's output but enter at reported survey values rather than being simulated from legislative rules. Because these instruments cannot support counterfactual decomposition (applying a different policy system to the same individual), I exclude them from all income components to avoid contaminating the decomposition with variation that cannot be cleanly attributed to policy rule changes. Their uniform uprating – governed since April 2011 by the State Pension triple lock (Thurley and Keen 2023) – also creates no cross-sectional variation in policy exposure beyond benefit receipt status. Second, policies outside UKMOD's scope entirely – indirect taxes (notably the 2011 VAT increase from 17.5% to 20%; HM Treasury 2010), Council Tax Support (localised to individual local authorities after April 2013; National Audit Office 2013; UK Parliament 2012), behavioural responses, and in-kind public services – are likewise not captured.

Simulated instruments capture the large majority of tax-and-benefit variation. When compared to the total tax-benefit signal (simulated plus non-simulated), simulated instruments account for 86% of cross-sectional variance and 83% of mean absolute changes.[16] For more details on the underlying reforms, see House of Commons Library (2016) on working-age benefit changes and Office for Budget Responsibility (2019) on welfare spending trends.

[16]The 2011-12 fiscal year is excluded from these coverage statistics because of the introduction of the State Pension triple lock (Thurley and Keen 2023) produced an anomalously large non-simulated signal for pensioners in that year. In all subsequent years, non-simulated policy changes are modest: less than £160/year for pensioners and less than £90/year for non-pensioners.

Table C1: Year-by-year timeline of major UK tax-and-transfer policy changes, 2010–11 to 2019–20, indicating which changes are included in the paper's simulated treatment variable.

Fiscal Year	Major Policy Changes	In simulated treatment?
2010–11	• Personal Allowance: £6,475 → £7,475 (Emergency Budget) • Higher-rate threshold lowered: £43,875 → £42,475 • Child Tax Credit family element: increased by £150 (above indexation) • VAT increase: 17.5% to 20% (Jan 2011; not in simulated treatment) • Standard policy parameter uprating for remaining instruments	✓(excl. VAT)
2011–12	• Personal Allowance: £7,475 • CPI uprating begins for benefits/tax credits • State Pension triple lock begins • Housing Benefit/LHA caps and percentile changes implemented • NI contributions: employee rates +1pp	✓(excl. State Pension)
2012–13	• Personal Allowance: £8,105 • Welfare Reform Act 2012 provisions begin implementation in subsequent years (UC, benefit cap, PIP, bedroom tax) • High Income Child Benefit Charge introduced (Jan 2013) • Contributory ESA time-limiting (WRAG) (contributory ESA not simulated from rules) • Additional rate 50% (cut to 45% announced for 2013–14)	✓(excl. contrib. ESA)
2013–14	• Personal Allowance: £9,440; Additional rate cut to 45% • "Bedroom tax" / under-occupancy deduction (Apr 2013) • Benefit cap introduced (£26k families, £18.2k singles) • Council Tax Benefit abolished, replaced by local Council Tax Support • UC pathfinder begins • PIP rollout begins (not simulated from rules) • 1% uprating cap for working-age benefits (2013–2016)	✓(excl. Council Tax, PIP)
2014–15	• Personal Allowance: £10,000 (coalition milestone) • 1% uprating cap continues • UC expansion continues • PIP reassessments expand (not simulated from rules)	✓(excl. PIP)

Continued on next page

Table C1 – continued from previous page

Fiscal Year	Major Policy Changes	In simulated treatment?
2015–16	• Personal Allowance: £10,600 • Marriage Allowance introduced	✓
2016–17	• Personal Allowance: £11,000 • Four-year benefit freeze begins (Apr 2016) • UC work allowance cuts implemented • Benefit cap lowered (Nov 2016) • New State Pension introduced (Apr 2016) • Personal Savings Allowance; dividend tax reforms	✓(excl. State Pension)
2017–18	• Personal Allowance: £11,500 • Two-child limit implemented (Apr 2017) • Removal of "family element" in tax credits/UC • UC taper rate cut (65% to 63%) • ESA WRAG component removed for new claimants (income-related ESA simulated; contributory ESA not) • Benefit freeze continues (2nd year)	✓(excl. contrib. ESA)
2018–19	• Personal Allowance: £11,850 • Scottish income tax diverges from rUK: starter rate 19%, intermediate rate 21% • Benefit freeze continues (3rd year) • UC rollout continues; legacy benefits closed to new claims	✓
2019–20	• Personal Allowance: £12,500; Higher-rate threshold: £50,000 • UC work allowances increased by £1,000/year (Apr 2019) • UC "SDP gateway" introduced (Jan 2019) • Benefit freeze continues (4th/final year)	✓(excl. State Pension)

Acronyms: VAT = Value Added Tax; RPI = Retail Prices Index; CPI = Consumer Prices Index; LHA = Local Housing Allowance; NI = National Insurance; ESA = Employment and Support Allowance; WRAG = Work-Related Activity Group; UC = Universal Credit; PIP = Personal Independence Payment; SDP = Severe Disability Premium.

D Validation of standard pocketbook voting relationships

This section validates that the standard pocketbook voting relationship holds in the linked UKMOD–UKHLS data. Figure D1 plots the coefficients from regressions of incumbent support on subjective pocketbook evaluations as well as on calculated total income changes, confirming that both predict incumbent support in the expected direction. Table D1 reports estimates from linear specifications treating each predictor as continuous. The figure instead uses categorical indicators for the pocketbook evaluation, which allows for nonlinearity in the relationship. As shown in Table D1, subjective pocketbook evaluations, raw income changes, and asinh-transformed income changes each significantly and positively predict incumbent support in lagged dependent variable models. Together, these results validate the full chain from calculated disposable income changes through subjective financial evaluations to incumbent vote choice.

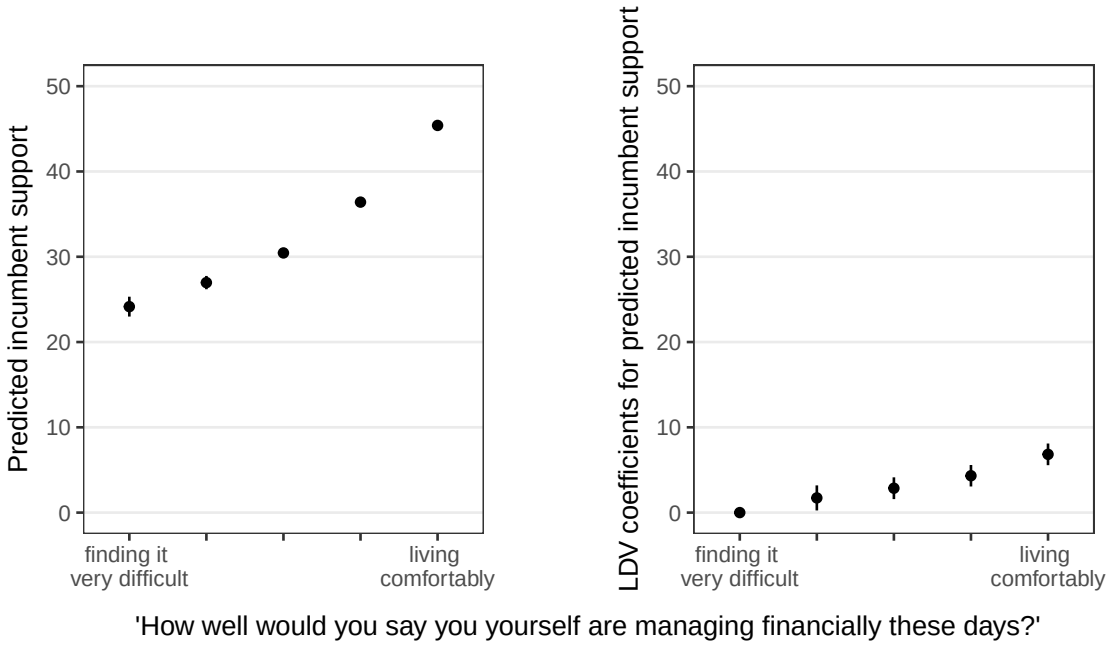


Figure D1: Regressions of lagged dependent variable models of incumbent support (0-100) with household-level clustered errors. Pocketbook evaluations are rescaled to a 0-100 scale. All income changes are yearly changes in monthly disposable household income (£), in real terms.

Table D1: Estimates from regressing incumbent support on pocketbook evaluations and calculated total income changes.

	1	2	3
Pocketbook evaluations	0.08 (0.00)***		
Δy^{total} (£1000/mo)		0.22 (0.06)***	
Δy^{total} (asinh)			0.08 (0.02)***
Con. (lag)	70.17 (0.33)***	71.43 (0.36)***	71.45 (0.36)***
Lab. (lag)	−5.54 (0.19)***	−5.13 (0.21)***	−5.13 (0.21)***
Lib Dem (lag)	37.41 (0.57)***	45.68 (0.61)***	45.68 (0.61)***
Employee (lag)	−1.70 (0.34)***	−1.68 (0.37)***	−1.71 (0.37)***
Pensioner (lag)	−1.73 (0.40)***	−1.41 (0.44)**	−1.51 (0.44)***
Unemployed (lag)	−5.15 (0.45)***	−6.70 (0.50)***	−6.81 (0.50)***
Student (lag)	−1.98 (0.56)***	−2.31 (0.65)***	−2.39 (0.64)***
Inactive (lag)	−3.53 (0.45)***	−3.93 (0.51)***	−4.01 (0.51)***
Sick/disabled (lag)	−5.61 (0.46)***	−7.20 (0.50)***	−7.27 (0.50)***
Other status (lag)	−1.43 (0.99)	−2.15 (1.14)	−2.25 (1.14)*
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Female	−0.44 (0.14)**	−0.16 (0.16)	−0.19 (0.16)
Age	0.10 (0.01)***	0.09 (0.01)***	0.09 (0.01)***
Num.Obs.	158 306	121 434	121 434
Std.Errors	Household	Household	Household
Lagged DV	✓	✓	✓

Validation models: total income change predicts both incumbent support and pocketbook evaluations. All income changes are yearly changes in monthly household disposable income (£), in real terms. Outcome: incumbent support or pocketbook evaluation (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table D2: Underlying estimates for Figure 2: mean income change by subjective financial evaluation change

	Income change (GBP/mo)
-3 steps	−7.66 (320.63)
-2 steps	289.83 (191.42)
-1 step	353.19 (190.50)
No change	407.71 (190.38)*
+1 step	464.88 (184.88)*
+2 steps	540.06 (192.73)**
+3 steps	661.61 (194.66)***
+4 steps	901.77 (225.43)***
Num.Obs.	161 040
Std.Errors	Household

OLS regression of yearly change in monthly disposable household income on factor-coded changes in subjective financial evaluations. Reference category: −4 steps (best to worst). Household-clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table D3: Underlying estimates for Figure D1: incumbent support by subjective financial evaluation

	Cross-sectional	LDV
Fin. eval.: 25	2.82 (0.65)***	1.72 (0.75)*
Fin. eval.: 50	6.30 (0.62)***	2.86 (0.65)***
Fin. eval.: 75	12.26 (0.63)***	4.32 (0.64)***
Fin. eval.: 100 (best)	21.25 (0.66)***	6.83 (0.65)***
Con. (lag) x Coal.		67.43 (5.55)***
Con. (lag) x Cons.		72.18 (5.56)***
Lab. (lag) x Coal.		−13.07 (5.55)*
Lab. (lag) x Cons.		−12.77 (5.54)*
Lib Dem (lag) x Coal.		62.22 (0.69)***
Lib Dem (lag) x Cons.		−1.20 (0.81)
Other (lag) x Coal.		−12.75 (5.55)*
Other (lag) x Cons.		−6.77 (5.56)
Party strength (lag)		3.52 (0.27)***
Disp. income (lag)		0.00 (0.00)**
Num.Obs.	288 839	91 951
Std.Errors	Individual	Individual
FE: wave		✓

Cross-sectional model regresses incumbent support (0–100) on factor-coded subjective financial evaluations; LDV model adds lagged party support interactions, party support strength, wave controls, and lagged disposable income. Reference category: 0 (finding it very difficult). Individual-clustered standard errors. * p<.05, ** p<.01, *** p<.001.

E Model specification

The main specification is a lagged dependent variable (LDV) model:

$$v_{it} = \beta_p \Delta y_{it}^{policy} + \beta_t \Delta y_{it}^{total} + \boldsymbol{p}'_{it-1}\boldsymbol{\gamma}_g + \boldsymbol{\mu}_{it} + \eta_t + \varepsilon_{it} \tag{4}$$

where v_{it} is incumbent support for individual i in fiscal year t, Δy_{it}^{policy} is the policy-induced change in average monthly disposable income (£) from the previous to the current fiscal year, Δy_{it}^{total} is the total income change, and η_t denotes wave fixed effects. $\boldsymbol{p}'_{it-1}\boldsymbol{\gamma}_g$ denotes interactions between lagged party support dummies (Conservative, Labour, Liberal Democrat, and other) and government period indicators (Labour 2010, Coalition 2010–2015, Conservative 2015–2019). Because "incumbent support" changes meaning when the governing party changes, a single lagged incumbent support term would conflate prior support for different parties. The interaction structure instead allows the model to flexibly condition on each respondent's prior partisan orientation within each government period. $\boldsymbol{\mu}_{it}$ includes controls: lagged employment status (as a factor), lagged real disposable income, lagged subjective pocketbook evaluations, gender, and age.

First, party support exhibits strong state dependence, i.e. successive measures are heavily conditioned on prior support, so a dynamic model that explicitly conditions on lagged partisanship is better specified than one that ignores this persistence (Tilley, Neundorf and Hobolt 2018). Second, individual fixed effects absorb only time-invariant heterogeneity, leaving the many potential time-varying confounders in this setting – shifting economic conditions, evolving policy environments, and changing personal circumstances – unaddressed.

As a robustness check, I also estimate two-way fixed effects (TWFE) models which address time-invariant unobserved heterogeneity directly by replacing the lagged party interactions with individual fixed effects α_i. Gender and age drop out of the TWFE specification because they are absorbed by α_i; the remaining time-varying controls are identical. The LDV and TWFE specifications rest on different and incompatible identifying assumptions: the LDV model assumes no time-invariant unobserved heterogeneity conditional on the lagged dependent variable and controls, while the TWFE model assumes strict exogeneity. As Ding and Li (2019) show, these two estimators bracket the causal parameter under their respective assumptions, so similar estimates across the two provides additional assurance that the results are not driven by the identifying assumptions of either specification alone.

In both specifications, β_p and β_t capture how incumbent support changes as a function of policy-induced vs. total income changes, holding the other constant. Attribution is estimated directly as the positive magnitude of β_p (equivalently, one can substitute $\Delta y_{it}^{nonpolicy}$ for Δy_{it}^{total}; see Appendix F). Under perfect policy-based attribution, β_p is positive and β_t is zero; under indiscriminate pocketbook voting, the reverse. In the LDV model, β_p captures the cross-sectional association between policy-induced income changes and incumbent support within each wave, conditional on the lagged party interactions and controls: a positive β_p means that voters who experienced larger policy-induced gains are more supportive of the incumbent than observationally similar voters in the same wave. With the inverse hyperbolic sine transformation, unit changes are approximately proportional rather than absolute.

Standard errors are clustered at the household level as described in the main text.

Table E1 reports the full model results underlying Table 1 in the main text, including all covariate estimates.

Table E1: Full model results for Table 1: policy-induced vs. total pocketbook changes.

	Total (raw)	Decomposed (raw)	Total (asinh)	Decomposed (asinh)
Δy^{total}	0.15 (0.03)***	0.16 (0.05)**	0.06 (0.02)**	0.06 (0.02)**
Δy^{policy}		−0.03 (0.15)		0.00 (0.03)
Con. (lag) x Coalition	66.26 (4.79)***	66.26 (4.79)***	66.25 (4.79)***	66.25 (4.79)***
Con. (lag) x Conservative	70.71 (4.83)***	70.71 (4.83)***	70.66 (4.82)***	70.67 (4.83)***
Lab. (lag) x Coalition	−9.32 (4.79)	−9.32 (4.79)	−9.36 (4.79)	−9.36 (4.79)
Lab. (lag) x Conservative	−7.83 (4.77)	−7.84 (4.77)	−7.89 (4.77)	−7.89 (4.77)
Lib Dem (lag) x Coalition	57.77 (0.69)***	57.77 (0.69)***	57.79 (0.69)***	57.79 (0.69)***
Lib Dem (lag) x Conservative	−0.62 (1.09)	−0.62 (1.09)	−0.64 (1.09)	−0.64 (1.09)
Other (lag) x Coalition	−8.85 (4.80)	−8.85 (4.80)	−8.89 (4.80)	−8.89 (4.80)
Other (lag) x Conservative	−2.55 (4.84)	−2.56 (4.84)	−2.62 (4.83)	−2.62 (4.84)
Employee (lag)	−2.03 (0.43)***	−2.03 (0.43)***	−2.05 (0.43)***	−2.05 (0.43)***
Pensioner (lag)	−1.91 (0.49)***	−1.91 (0.50)***	−1.96 (0.49)***	−1.96 (0.50)***
Unemployed (lag)	−5.15 (0.64)***	−5.15 (0.64)***	−5.24 (0.64)***	−5.24 (0.64)***
Student (lag)	−2.94 (0.79)***	−2.93 (0.79)***	−3.01 (0.79)***	−3.01 (0.79)***
Inactive (lag)	−2.61 (0.62)***	−2.61 (0.62)***	−2.67 (0.62)***	−2.67 (0.62)***
Sick/disabled (lag)	−5.16 (0.62)***	−5.15 (0.62)***	−5.19 (0.62)***	−5.19 (0.62)***
Other status (lag)	0.04 (1.38)	0.05 (1.38)	0.00 (1.38)	0.00 (1.39)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***
Female	−0.54 (0.19)**	−0.54 (0.19)**	−0.56 (0.19)**	−0.56 (0.19)**
Age	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***
Num.Obs.	91 295	91 295	91 295	91 295
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Full model results underlying Table 1. Total = policy-induced + non-policy. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

F Robustness to model specification

This section examines the sensitivity of the main results to model specification choices along three dimensions: alternative estimation strategies, additional controls and sample restrictions, and progressive control addition. Together, the three tables below show that the core finding – no differential effect of policy-induced income changes on incumbent support – is not an artefact of any particular modelling decision.

Table F1 varies the estimation strategy. All models use the main treatment variable for policy-induced income changes. The model in column 1 shows a fully decomposed model using raw income changes (£ 1000/mo), while the model in column 2 uses the inverse hyperbolic sine transformation. The model in column 3 is a 'naive' pooled cross-sectional model pooling all observations without unit or wave fixed effects, lagged dependent variables, or controls. The models in columns 4 and 5 show versions using unit fixed effects or two-way fixed effects (with controls). The model in column 6 shows the main specification but with income changes at the individual rather than household level.

The TWFE estimates (column 5) agree with the main estimates on the key quantity of interest: the differential policy effect is null or modest under both estimators, providing assurance that the attribution finding is not driven by the identifying assumptions of either specification alone. The unit fixed effects specification (column 4) shows a marginally significant positive differential for policy-induced changes, but this attenuates and loses significance when wave fixed effects are added (column 5). This pattern is consistent with the unit-FE estimate partly capturing aggregate time variation – since policy rule changes are largely common shocks within each fiscal year, unit fixed effects alone cannot separate individual-level policy exposure from other year-specific influences on incumbent support. The TWFE specification, which absorbs these common shocks, provides the more appropriate comparison to the LDV results.

Table F2 shows results for further alternative specifications that add variables or exclude observations. They all use the main model specification with lagged dependent variables and controls. All models use the inverse hyperbolic sine transformation except Model 2, which uses raw income changes (£ 1000/mo). Model 1 includes dummies for the most extreme 10 percent of positive and negative income changes (as in Tilley, Neundorf and Hobolt (2018)). Model 2 includes squared terms. Model 3 includes lagged versions of the total and policy-induced income variables. Model 4 excludes the top decile of income earners, i.e. those who have an average disposable income in the top decile in the sample. Model 5 addresses potential benefit non-take-up by excluding all observations for households whose real benefit income ever falls at or below the 25th percentile of positive transfers (pooled across all person-years), for whom the risk of non-take-up is greatest. Because this threshold is computed among positive values, households with zero benefit income in any year are also excluded. This rule applied across the panel substantially reduces the sample (to 21,039 observations). The differential policy effect remains null in this restricted sample.

Finally, Table F3 documents how the estimated coefficients on total income change and the differential policy effect vary as the control set expands. Columns add controls progressively – from no controls, through basic demographic and economic controls, to the full specification including party-strength interactions and job loss/gain dummies – revealing the sensitivity of the treatment effect estimates to the control set. The Full LDV column adds lagged party-support-strength interactions; the Extra controls column further adds job loss and job gain indicators. The two columns yield identical estimates at reported precision, confirming that job transitions do not

Table F1: Robustness: Specification variants

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta y^{nonpolicy}$	0.19 (0.05)***	0.10 (0.02)***				
Δy^{total}			0.12 (0.02)***	−0.03 (0.02)	−0.01 (0.02)	
Δy^{policy}	0.19 (0.12)	0.04 (0.03)	0.18 (0.04)***	0.07 (0.03)*	0.05 (0.03)	
Δy^{total} (ind.)						0.10 (0.02)***
Δy^{policy} (ind.)						−0.03 (0.03)
Con. (lag) x Coal.	67.28 (4.78)***	67.31 (4.77)***				66.26 (4.88)***
Con. (lag) x Cons.	70.98 (4.80)***	70.96 (4.80)***				70.64 (4.91)***
Lab. (lag) x Coal.	−8.28 (4.77)	−8.29 (4.77)				−9.37 (4.87)
Lab. (lag) x Cons.	−7.45 (4.75)	−7.46 (4.75)				−7.94 (4.86)
Lib Dem (lag) x Coal.	57.64 (0.69)***	57.67 (0.69)***				57.80 (0.69)***
Lib Dem (lag) x Cons.	−1.36 (1.09)	−1.39 (1.09)				−0.59 (1.10)
Other (lag) x Coal.	−7.73 (4.78)	−7.74 (4.77)				−8.88 (4.88)
Other (lag) x Cons.	−1.68 (4.81)	−1.70 (4.81)				−2.66 (4.92)
Employee (lag)	−2.00 (0.43)***	−2.03 (0.43)***		−0.81 (0.87)	−0.94 (0.86)	−2.04 (0.43)***
Pensioner (lag)	−1.90 (0.50)***	−1.96 (0.50)***		−0.55 (1.11)	0.10 (1.11)	−2.01 (0.50)***
Unemployed (lag)	−5.14 (0.64)***	−5.25 (0.64)***		−2.06 (1.04)*	−2.33 (1.04)*	−5.36 (0.64)***
Student (lag)	−2.93 (0.79)***	−3.02 (0.79)***		−0.33 (1.43)	−0.94 (1.43)	−3.04 (0.79)***
Inactive (lag)	−2.57 (0.62)***	−2.64 (0.62)***		−0.97 (1.12)	−1.18 (1.12)	−2.72 (0.62)***
Sick/disabled (lag)	−5.11 (0.63)***	−5.17 (0.63)***		−1.34 (1.21)	−1.35 (1.21)	−5.16 (0.63)***
Other status (lag)	0.04 (1.38)	−0.04 (1.39)		−0.77 (1.61)	−0.88 (1.61)	−0.04 (1.39)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***		0.00 (0.00)	0.00 (0.00)	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***		0.01 (0.01)	0.02 (0.01)*	0.06 (0.00)***
Female	−0.54 (0.19)**	−0.57 (0.19)**				−0.52 (0.19)**
Age	0.08 (0.01)***	0.08 (0.01)***				0.08 (0.01)***
Num.Obs.	91 295	91 295	141 222	136 884	136 884	91 127
Std.Errors	Household	Household	Household	Household	Household	Household
FE: wave					✓	✓
FE: Individual				✓	✓	
Lagged DV	✓	✓				✓

Robustness to model specification. Columns span pooled OLS, unit fixed effects, two-way fixed effects, and LDV variants. Policy effect is simulated-only (rule changes). All income changes are yearly changes in monthly household disposable income (£), in real terms; asinh-transformed except where noted. Outcome: incumbent support (0–100). Standard errors are household-clustered; see FE rows for fixed-effects structure. * p<.05, ** p<.01, *** p<.001.

Table F2: Robustness summary: Combined checks

	(1)	(2)	(3)	(4)	(5)
Δy^{total}	0.04 (0.02)	0.30 (0.06)***	0.10 (0.02)***	0.05 (0.02)**	0.12 (0.04)**
Δy^{policy}	−0.03 (0.03)	−0.07 (0.17)	0.03 (0.03)	0.02 (0.03)	0.05 (0.05)
Δy^{total} extreme (bottom 10%)	0.52 (0.45)				
Δy^{total} extreme (top 10%)	2.13 (0.40)***				
Δy^{total}_{t-1}			0.08 (0.02)***		
Δy^{policy}_{t-1}			−0.03 (0.03)		
Con. (lag) x Coal.	67.22 (4.78)***	67.26 (4.78)***	65.76 (4.87)***	67.34 (4.92)***	56.72 (9.64)***
Con. (lag) x Cons.	70.97 (4.80)***	70.98 (4.80)***	69.50 (4.89)***	71.82 (4.96)***	62.72 (9.68)***
Lab. (lag) x Coal.	−8.30 (4.77)	−8.28 (4.77)	−10.38 (4.86)*	−7.21 (4.92)	−20.84 (9.63)*
Lab. (lag) x Cons.	−7.43 (4.75)	−7.42 (4.75)	−9.35 (4.84)	−6.13 (4.89)	−21.27 (9.61)*
Lib Dem (lag) x Coal.	57.64 (0.69)***	57.63 (0.69)***	60.07 (0.84)***	57.00 (0.74)***	60.31 (1.50)***
Lib Dem (lag) x Cons.	−1.38 (1.09)	−1.37 (1.09)	−1.42 (1.13)	−1.21 (1.17)	−2.28 (2.52)
Other (lag) x Coal.	−7.77 (4.78)	−7.73 (4.78)	−9.70 (4.87)*	−6.69 (4.93)	−19.75 (9.64)*
Other (lag) x Cons.	−1.68 (4.81)	−1.66 (4.81)	−3.51 (4.89)	−0.52 (4.96)	−11.66 (9.68)
Employee (lag)	−1.79 (0.43)***	−2.03 (0.43)***	−1.71 (0.49)***	−2.09 (0.47)***	−2.39 (1.79)
Pensioner (lag)	−1.80 (0.50)***	−1.89 (0.50)***	−1.72 (0.57)**	−1.54 (0.55)**	−1.77 (1.63)
Unemployed (lag)	−5.09 (0.64)***	−5.10 (0.64)***	−4.83 (0.79)***	−5.03 (0.67)***	−5.07 (1.85)**
Student (lag)	−2.89 (0.79)***	−2.86 (0.79)***	−3.72 (0.98)***	−2.50 (0.83)**	−5.04 (3.52)
Inactive (lag)	−2.46 (0.62)***	−2.49 (0.63)***	−1.57 (0.73)*	−2.60 (0.66)***	−3.33 (1.94)
Sick/disabled (lag)	−4.93 (0.63)***	−5.07 (0.63)***	−5.15 (0.76)***	−4.89 (0.65)***	−3.89 (1.75)*
Other status (lag)	0.10 (1.39)	0.09 (1.39)	1.23 (1.53)	0.03 (1.46)	−0.32 (4.02)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)**	0.00 (0.00)***	0.00 (0.00)**
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.01)***	0.05 (0.00)***	0.06 (0.01)***
Female	−0.58 (0.19)**	−0.49 (0.19)*	−0.81 (0.22)***	−0.29 (0.21)	0.17 (0.38)
Age	0.07 (0.01)***	0.08 (0.01)***	0.07 (0.01)***	0.08 (0.01)***	0.07 (0.02)**
Num.Obs.	91 295	91 295	62 902	81 756	21 039
Std.Errors	Household	Household	Household	Household	Household
Lagged DV	✓	✓	✓	✓	✓
Wave FE	✓	✓	✓	✓	✓

Combined robustness table. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Column (1) adds indicators for the 10% most extreme income changes. Column (2) uses raw income changes (£1k/month) with squared terms. Column (3) adds lagged income changes. Column (4) excludes the top income decile. Column (5) excludes households ever in the lowest quartile of positive transfers. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table F3: Robustness: Control-set sensitivity

	No controls	Basic controls	Simple LDV	Full LDV	Extra controls
Δy^{total}	0.12 (0.02)***	0.15 (0.02)***	0.08 (0.02)***	0.05 (0.02)*	0.05 (0.02)*
Δy^{policy}	0.18 (0.04)***	−0.14 (0.04)***	−0.03 (0.03)	0.01 (0.03)	0.01 (0.03)
Con. (lag) x Coal.				60.63 (5.41)***	60.59 (5.40)***
Con. (lag) x Cons.				65.66 (5.43)***	65.62 (5.43)***
Lab. (lag) x Coal.				−14.93 (5.40)**	−14.96 (5.40)**
Lab. (lag) x Cons.				−13.95 (5.39)**	−13.99 (5.38)**
Lib Dem (lag) x Coal.				58.88 (0.84)***	58.88 (0.84)***
Lib Dem (lag) x Cons.				−5.99 (1.35)***	−5.99 (1.35)***
Other (lag) x Coal.				−15.43 (5.41)**	−15.47 (5.41)**
Other (lag) x Cons.				−9.20 (5.44)	−9.23 (5.44)
Incumbent support (lag)			0.66 (0.00)***		
Party strength (lag)				−3.16 (0.66)***	−3.16 (0.66)***
Con. (lag) x Strength				18.70 (0.96)***	18.70 (0.96)***
Lab. (lag) x Strength				−2.28 (0.73)**	−2.29 (0.73)**
Lib Dem (lag) x Strength				24.89 (2.03)***	24.87 (2.03)***
Employee (lag)		−7.40 (0.81)***	−2.37 (0.40)***	−1.64 (0.44)***	−1.64 (0.44)***
Pensioner (lag)		−5.85 (1.01)***	−2.22 (0.47)***	−2.09 (0.51)***	−2.09 (0.51)***
Unemployed (lag)		−15.59 (0.95)***	−5.93 (0.53)***	−4.52 (0.73)***	−5.09 (0.92)***
Student (lag)		−8.77 (1.18)***	−3.76 (0.67)***	−2.27 (0.88)**	−2.18 (0.89)*
Inactive (lag)		−11.99 (1.03)***	−3.81 (0.53)***	−3.24 (0.68)***	−3.16 (0.68)***
Sick/disabled (lag)		−19.16 (1.02)***	−6.49 (0.52)***	−4.40 (0.70)***	−4.31 (0.70)***
Other status (lag)		−8.06 (1.82)***	−2.21 (1.14)	0.55 (1.52)	0.74 (1.53)
Disp. income (lag)		0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)		0.22 (0.01)***	0.07 (0.00)***	0.05 (0.00)***	0.05 (0.00)***
Female		−0.97 (0.34)**	−0.53 (0.16)***	−0.27 (0.21)	−0.28 (0.21)
Age		0.28 (0.02)***	0.09 (0.01)***	0.05 (0.01)***	0.05 (0.01)***
Job loss					−1.72 (0.96)
Job gain					0.92 (1.12)
Num.Obs.	141 222	136 884	121 366	70 772	70 772
Std.Errors	Household	Household	Household	Household	Household
FE: wave		✓	✓	✓	✓
Lagged DV			✓	✓	✓

Robustness to control specification. The main specification includes lagged party support interacted with government dummies; simpler specifications omit these interactions. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

affect the results. The reduction in sample size from the Simple LDV to the Full LDV specification reflects additional missingness in the lagged party-support-strength measure. Note that none of these specifications are identical to the main specification reported in the body of the paper: the "Full" column includes job loss and job gain indicators that are not part of the main model, making this table strictly a sensitivity exercise showing how results change as the control set is progressively saturated. This transparency table documents the full specification path rather than reporting only the preferred specification.

Table G1: Robustness: Weighted vs unweighted estimates

	Unweighted (main)	Weighted
Δy^{total}	0.08 (0.02)***	0.09 (0.03)***
Δy^{policy}	0.01 (0.03)	−0.01 (0.04)
Con. (lag) x Coalition	67.28 (4.77)***	58.50 (3.95)***
Con. (lag) x Conservative	70.97 (4.80)***	62.48 (4.00)***
Lab. (lag) x Coalition	−8.31 (4.77)	−16.62 (3.93)***
Lab. (lag) x Conservative	−7.47 (4.75)	−15.55 (3.91)***
Lib Dem (lag) x Coalition	57.66 (0.69)***	55.57 (0.94)***
Lib Dem (lag) x Conservative	−1.39 (1.09)	−0.73 (1.53)
Other (lag) x Coalition	−7.76 (4.78)	−14.27 (3.95)***
Other (lag) x Conservative	−1.70 (4.81)	−9.56 (3.99)*
Employee (lag)	−2.03 (0.43)***	−2.12 (0.60)***
Pensioner (lag)	−1.97 (0.50)***	−1.92 (0.68)**
Unemployed (lag)	−5.26 (0.64)***	−6.47 (0.90)***
Student (lag)	−3.02 (0.79)***	−3.00 (1.16)**
Inactive (lag)	−2.65 (0.62)***	−2.32 (0.88)**
Sick/disabled (lag)	−5.16 (0.63)***	−6.01 (0.85)***
Other status (lag)	−0.02 (1.38)	−1.44 (1.79)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.01)***
Female	−0.56 (0.19)**	−0.51 (0.28)
Age	0.08 (0.01)***	0.08 (0.01)***
Num.Obs.	91 295	60 469
Std.Errors	Household	Household
Lagged DV	✓	✓
Wave FE	✓	✓

Comparison of unweighted and weighted estimates using UKHLS longitudinal weights. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

G Weighted estimates

The main analysis uses unweighted estimates to maximize sample size and maintain consistency with the microsimulation procedure, which does not apply survey weights. This section replicates the main specification using UKHLS longitudinal weights (`indpxus_lw` for waves 2+, `indpxus_xw` for wave 1) to verify that results are not sensitive to sample composition adjustments. As Table G1 shows, the weighted estimates are substantively identical to the unweighted main results.

Table H1: Robustness: Stable household-composition spells

	Total (raw)	Decomposed (raw)	Total (asinh)	Decomposed (asinh)
Δy^{total}	0.14 (0.03)***	0.16 (0.05)**	0.06 (0.02)**	0.06 (0.02)**
Δy^{policy}		−0.05 (0.15)		−0.01 (0.03)
Con. (lag) x Coalition	65.62 (4.83)***	65.62 (4.83)***	65.62 (4.83)***	65.62 (4.83)***
Con. (lag) x Conservative	70.16 (4.86)***	70.15 (4.86)***	70.11 (4.86)***	70.10 (4.86)***
Lab. (lag) x Coalition	−10.13 (4.82)*	−10.13 (4.82)*	−10.16 (4.82)*	−10.16 (4.82)*
Lab. (lag) x Conservative	−8.50 (4.81)	−8.51 (4.81)	−8.56 (4.81)	−8.57 (4.81)
Lib Dem (lag) x Coalition	57.90 (0.70)***	57.90 (0.70)***	57.92 (0.70)***	57.92 (0.70)***
Lib Dem (lag) x Conservative	−0.67 (1.12)	−0.67 (1.12)	−0.68 (1.12)	−0.68 (1.12)
Other (lag) x Coalition	−9.60 (4.83)*	−9.60 (4.83)*	−9.63 (4.83)*	−9.63 (4.83)*
Other (lag) x Conservative	−3.24 (4.87)	−3.25 (4.87)	−3.30 (4.87)	−3.32 (4.87)
Employee (lag)	−2.05 (0.44)***	−2.05 (0.44)***	−2.07 (0.44)***	−2.07 (0.44)***
Pensioner (lag)	−2.04 (0.50)***	−2.03 (0.50)***	−2.08 (0.50)***	−2.08 (0.50)***
Unemployed (lag)	−5.07 (0.66)***	−5.07 (0.66)***	−5.15 (0.66)***	−5.15 (0.66)***
Student (lag)	−2.60 (0.83)**	−2.59 (0.83)**	−2.67 (0.83)**	−2.66 (0.83)**
Inactive (lag)	−2.49 (0.64)***	−2.48 (0.64)***	−2.55 (0.64)***	−2.55 (0.64)***
Sick/disabled (lag)	−5.30 (0.64)***	−5.29 (0.64)***	−5.33 (0.64)***	−5.33 (0.64)***
Other status (lag)	−0.76 (1.43)	−0.76 (1.44)	−0.80 (1.44)	−0.80 (1.44)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***
Female	−0.54 (0.20)**	−0.54 (0.20)**	−0.56 (0.20)**	−0.56 (0.20)**
Age	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***
Num.Obs.	88001	88001	88001	88001
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Estimated on stable household-composition spells (at least 2 consecutive observed waves). Composition defined by exact household membership; spells are split at membership changes and observation gaps. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors (clustered at the original household panel identifier). * p<.05, ** p<.01, *** p<.001.

H Robustness to household composition changes

The main analysis uses household-level income variables measured at the household identifier level. Changes in household composition between survey waves – such as individuals moving in or out of the household – could mechanically affect measured income changes and confound the estimated relationship between policy-induced income changes and incumbent support. This section addresses this concern by restricting the sample to stable household-composition spells, defined as sequences of at least two consecutive observed waves in which exact household membership does not change. Spells are split at membership changes and observation gaps, so a household that experiences a composition change contributes separate spells before and after. Results in Table H1 are substantively similar to the main specification, indicating that household composition changes do not drive the findings.

I Robustness to sample period restrictions

The main analysis pools all available waves of the Understanding Society panel (2009–2019). This section tests whether results are sensitive to the time period covered, first by excluding waves affected by the Brexit referendum and then by examining effects separately across governing periods. Both analyses suggest that the findings are not driven by any particular sub-period.

Excluding post-referendum waves

The Brexit referendum in June 2016 may have fundamentally altered the relationship between economic conditions and political attitudes. While the policy decomposition itself is mechanical and unaffected by macroeconomic conditions, the noise component (non-policy income changes) and political attitudes may both have been influenced by Brexit-related uncertainty. This table restricts the sample to waves before the referendum period (excluding 2017–2019). Because 2016 is absent from the data due to missing simulation output, and the LDV specification requires a lagged observation, 2017 observations are already non-estimable regardless of the exclusion. The effective sample reduction therefore reflects only the loss of 2018 and 2019 observations. As Table I1 shows, results are substantively similar to the full-sample estimates.

Table I1: Robustness: Excluding Brexit years (2017-2019)

	Total (raw)	Decomposed (raw)	Total (asinh)	Decomposed (asinh)
Δy^{total}	0.15 (0.03)***	0.16 (0.05)**	0.06 (0.02)**	0.06 (0.02)**
Δy^{policy}		−0.03 (0.15)		−0.01 (0.03)
Con. (lag) x Coalition	66.69 (4.85)***	66.69 (4.85)***	66.69 (4.84)***	66.69 (4.84)***
Con. (lag) x Conservative	72.06 (4.88)***	72.06 (4.88)***	72.03 (4.88)***	72.01 (4.88)***
Lab. (lag) x Coalition	−8.88 (4.84)	−8.88 (4.84)	−8.91 (4.84)	−8.91 (4.84)
Lab. (lag) x Conservative	−8.21 (4.82)	−8.21 (4.82)	−8.26 (4.82)	−8.29 (4.82)
Lib Dem (lag) x Coalition	57.76 (0.69)***	57.76 (0.69)***	57.78 (0.69)***	57.78 (0.69)***
Lib Dem (lag) x Conservative	−0.08 (1.28)	−0.08 (1.28)	−0.11 (1.29)	−0.11 (1.29)
Other (lag) x Coalition	−8.41 (4.85)	−8.41 (4.85)	−8.44 (4.85)	−8.44 (4.85)
Other (lag) x Conservative	−2.25 (4.89)	−2.26 (4.89)	−2.31 (4.89)	−2.33 (4.89)
Employee (lag)	−2.07 (0.43)***	−2.07 (0.43)***	−2.09 (0.43)***	−2.09 (0.43)***
Pensioner (lag)	−2.05 (0.50)***	−2.04 (0.50)***	−2.10 (0.50)***	−2.09 (0.50)***
Unemployed (lag)	−5.18 (0.65)***	−5.18 (0.65)***	−5.28 (0.66)***	−5.27 (0.66)***
Student (lag)	−2.99 (0.80)***	−2.99 (0.80)***	−3.07 (0.80)***	−3.06 (0.80)***
Inactive (lag)	−2.53 (0.63)***	−2.53 (0.63)***	−2.60 (0.63)***	−2.60 (0.63)***
Sick/disabled (lag)	−5.10 (0.63)***	−5.10 (0.63)***	−5.14 (0.63)***	−5.13 (0.63)***
Other status (lag)	0.21 (1.43)	0.22 (1.43)	0.17 (1.43)	0.17 (1.43)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***
Female	−0.49 (0.19)*	−0.48 (0.19)*	−0.50 (0.19)**	−0.50 (0.19)**
Age	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***
Num.Obs.	88074	88074	88074	88074
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Pre-Brexit sample: excludes 2017–2019 (post-referendum); 2016 is absent due to missing simulation output. Total = policy-induced + non-policy. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table I2: Governing period heterogeneity: Policy-induced pocketbook changes relative to non-policy changes

	Labour (policy-induced only)	Coalition (decomp.)	Conservatives (decomp.)
Δy^{total}		0.07 (0.02)***	0.04 (0.03)
Δy^{policy}	0.08 (0.33)	−0.02 (0.03)	0.01 (0.29)
Con. (lag) x Coalition		275.78 (0.78)***	14.05 (1.06)***
Con. (lag) x Conservative		17.24 (0.94)***	73.40 (0.77)***
Lab. (lag) x Coalition		200.20 (0.43)***	−2.44 (3.57)
Lab. (lag) x Conservative		−1.61 (3.49)	−5.24 (0.64)***
Lib Dem (lag) x Coalition		57.78 (0.69)***	3.94 (28.05)
Lib Dem (lag) x Conservative		−53.32 (27.31)	−0.41 (1.10)
Other (lag) x Coalition		200.69 (0.49)***	11.70 (15.71)
Other (lag) x Conservative		16.89 (15.97)	
Employee (lag)		−2.25 (0.48)***	−1.38 (0.92)
Pensioner (lag)		−2.41 (0.56)***	0.21 (1.05)
Unemployed (lag)		−5.47 (0.72)***	−3.77 (1.45)**
Student (lag)		−2.40 (0.89)**	−5.34 (1.78)**
Inactive (lag)		−2.30 (0.71)**	−4.54 (1.48)**
Sick/disabled (lag)		−5.11 (0.70)***	−5.46 (1.56)***
Other status (lag)		0.70 (1.61)	−3.17 (2.64)
Disp. income (lag)		0.00 (0.00)***	0.00 (0.00)
Pocketbook eval. (lag)		0.06 (0.01)***	0.05 (0.01)***
Female	0.58 (1.44)	−0.31 (0.22)	−1.35 (0.43)**
Age	0.11 (0.06)*	0.08 (0.01)***	0.05 (0.02)*
Num.Obs.	2606	74396	16958
Std.Errors	Household	Household	Household
FE: wave		✓	✓
Lagged DV		✓	✓

Labor model uses basic demographic controls only; other models use the main specification. Total = policy-induced + non-policy. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Effects by governing period

Table I2 shows results split by the three governing periods covered by the data: Labour (2010), Coalition (2010–2015), and Conservatives (2015–2019). Because Labour is observed in only one wave (2010), a lagged dependent variable model is not possible for that period; the first column therefore reports a cross-sectional between-individual model with basic demographic controls and only Δy^{policy}, since Δy^{total} requires market incomes and demographics from the previous survey wave. The remaining two columns estimate the main LDV specification for the Coalition and Conservative periods. The Labour coefficient is imprecisely estimated and not statistically significant. For the Coalition and Conservative periods, the pattern is consistent with the main results: total income changes matter in the expected direction during the Coalition years, but the policy-specific differential is small and insignificant across both periods, suggesting that the party in power makes little difference.

J Heterogeneous effects

This appendix supplements the heterogeneity analysis reported in the main text (see Figure 5). All models use the LDV specification in Equation 4 with the inverse hyperbolic sine transformation. For continuous moderators (age, political interest, average income, policy-induced share), marginal effects from a linear interaction model are plotted across the moderator's observed range, with 95% confidence intervals. For categorical or binary moderators (lagged vote, economic status), separate regressions are estimated for each group. Table J1 reports the continuous-moderator interaction models and the lagged incumbent-support subgroup models. Table J2 reports the economic status subgroup models separately. Together, these correspond to the heterogeneity analysis discussed in the main text.

Table J1: Heterogeneity models: Continuous moderators and lagged incumbent-support groups

	Age	Political interest	Avg. income	Policy-induced share	Lagged support: non-incumbent	Lagged support: incumbent
Δy^{total}	0.06 (0.02)**	0.05 (0.02)**	0.05 (0.02)*	0.05 (0.02)**	0.02 (0.02)	0.12 (0.04)**
Δy^{policy}	−0.11 (0.09)	−0.02 (0.05)	0.02 (0.04)	−0.06 (0.05)	0.03 (0.03)	−0.08 (0.05)
Age	0.08 (0.01)***	0.07 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	−0.01 (0.01)	0.22 (0.02)***
Mean political interest		7.79 (0.36)***				
Mean disp. income			0.00 (0.00)**			
Policy share of Δy				−0.62 (0.47)		
Δy^{policy} × Age	0.00 (0.00)					
Δy^{policy} × Mean political interest		0.06 (0.08)				
Δy^{policy} × Mean disp. income			0.00 (0.00)			
Δy^{policy} × Policy share of Δy				0.20 (0.10)*		
Con. (lag) x Coalition	66.23 (4.79)***	66.26 (4.79)***	66.27 (4.80)***	66.21 (4.79)***	79.36 (4.39)***	22.91 (8.69)**
Con. (lag) x Conservative	70.68 (4.83)***	70.47 (4.82)***	70.67 (4.83)***	70.66 (4.82)***		27.39 (8.70)**
Lab. (lag) x Coalition	−9.37 (4.79)	−9.27 (4.78)	−9.27 (4.79)	−9.38 (4.78)*	0.58 (3.99)	−50.62 (8.81)***
Lab. (lag) x Conservative	−7.89 (4.77)	−8.01 (4.77)	−7.80 (4.78)	−7.90 (4.77)	1.72 (3.98)	
Lib Dem (lag) x Coalition	57.78 (0.69)***	57.46 (0.69)***	57.73 (0.69)***	57.76 (0.69)***	43.89 (3.38)***	7.03 (8.71)
Lib Dem (lag) x Conservative	−0.64 (1.09)	−1.20 (1.10)	−0.77 (1.10)	−0.67 (1.09)	−4.16 (1.54)**	−46.30 (8.78)***
Other (lag) x Coalition	−8.90 (4.80)	−8.39 (4.79)	−8.77 (4.80)	−8.90 (4.79)	1.31 (3.99)	
Other (lag) x Conservative	−2.62 (4.84)	−2.33 (4.83)	−2.48 (4.84)	−2.60 (4.83)	7.57 (4.05)	
Employee (lag)	−2.04 (0.43)***	−1.80 (0.43)***	−2.00 (0.43)***	−2.05 (0.43)***	−0.73 (0.48)	−3.08 (0.78)***
Pensioner (lag)	−1.98 (0.50)***	−1.57 (0.50)**	−1.68 (0.50)***	−1.95 (0.50)***	0.04 (0.56)	−4.52 (0.89)***
Unemployed (lag)	−5.23 (0.64)***	−4.61 (0.64)***	−4.95 (0.65)***	−5.20 (0.64)***	−2.47 (0.59)***	−15.19 (2.02)***
Student (lag)	−3.01 (0.79)***	−3.26 (0.79)***	−2.62 (0.79)***	−3.21 (0.79)***	−1.11 (0.77)	−7.33 (2.11)***
Inactive (lag)	−2.68 (0.62)***	−2.17 (0.62)***	−2.30 (0.63)***	−2.63 (0.62)***	−1.18 (0.63)	−5.92 (1.55)***
Sick/disabled (lag)	−5.18 (0.62)***	−4.41 (0.63)***	−4.92 (0.63)***	−5.20 (0.63)***	−2.51 (0.60)***	−13.16 (2.23)***
Other status (lag)	0.00 (1.38)	0.17 (1.37)	0.29 (1.38)	0.04 (1.39)	−0.78 (1.37)	1.72 (2.91)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)	0.00 (0.00)***	0.00 (0.00)	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.05 (0.00)***	0.05 (0.00)***	0.06 (0.00)***	0.03 (0.00)***	0.11 (0.01)***
Female	−0.56 (0.19)**	0.20 (0.19)	−0.34 (0.21)	−0.55 (0.19)**	−0.30 (0.20)	−1.12 (0.44)*
Num.Obs.	91295	91294	91295	91244	54450	36845
Std.Errors	Household	Household	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓	✓	✓

Models underlying the heterogeneity plot for continuous moderators and lagged incumbent-support groups. Continuous-moderator panels use interaction models; lagged incumbent support panels use subgroup regressions. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table J2: Heterogeneity models: Economic status groups (LES)

	LES 3: Self-emp.	LES 4: Employee	LES 5: Pensioner	LES 6: Unemployed	LES 7: Student	LES 8: Inactive	LES 9: Sick/disabled	LES 10: Other
Δy^{total}	0.12 (0.06)	0.04 (0.03)	0.06 (0.03)	−0.02 (0.10)	0.01 (0.14)	0.08 (0.09)	0.18 (0.09)*	0.13 (0.24)
Δy^{policy}	−0.04 (0.09)	−0.05 (0.05)	−0.01 (0.04)	−0.02 (0.12)	−0.16 (0.25)	−0.01 (0.12)	−0.05 (0.12)	0.47 (0.33)
Age	0.15 (0.04)***	0.08 (0.01)***	0.00 (0.02)	0.02 (0.04)	0.06 (0.12)	0.16 (0.05)***	0.10 (0.05)	0.25 (0.11)*
Con. (lag) x Coalition	69.09 (4.61)***	72.23 (7.05)***	61.57 (5.71)***	53.57 (3.62)***	32.99 (17.14)	64.55 (2.79)***	52.88 (3.84)***	70.58 (7.33)***
Con. (lag) x Conservative	73.10 (4.86)***	75.05 (7.12)***	68.19 (5.75)***	61.11 (6.39)***	30.15 (17.89)	54.25 (5.20)***	57.60 (6.29)***	60.40 (12.21)***
Lab. (lag) x Coalition	−9.15 (4.51)*	−2.10 (7.05)	−18.17 (5.69)**	−3.15 (2.56)	−30.33 (16.96)	−3.60 (1.79)*	−9.13 (2.68)***	−8.67 (4.64)
Lab. (lag) x Conservative	−7.39 (4.68)	−0.35 (7.02)	−16.85 (5.66)**	0.97 (2.72)	−33.13 (17.12)	−6.82 (1.72)***	−7.04 (3.34)*	−11.70 (7.27)
Lib Dem (lag) x Coalition	57.68 (2.30)***	57.02 (0.96)***	61.22 (1.20)***	50.92 (3.79)***	47.31 (4.03)***	54.80 (3.27)***	52.74 (4.47)***	44.70 (8.55)***
Lib Dem (lag) x Conservative	−2.47 (3.37)	3.46 (1.63)*	−4.65 (1.87)*	1.47 (5.29)	−15.56 (4.54)***	−0.64 (5.29)	−7.07 (2.56)**	−4.00 (15.98)
Other (lag) x Coalition	−7.41 (4.51)	−2.46 (7.06)	−15.50 (5.71)**	−4.38 (2.48)	−32.09 (16.95)	−5.94 (2.10)**	−10.31 (2.51)***	−7.40 (5.73)
Other (lag) x Conservative	−0.27 (4.72)	2.22 (7.13)	−6.98 (5.79)		−26.05 (17.58)	−4.31 (2.91)	−2.74 (2.10)	4.31 (12.69)
Employee (lag)	−0.91 (1.17)	−2.28 (1.10)*	−1.43 (2.00)	−0.17 (3.96)	−6.65 (8.09)	−8.34 (4.77)	−4.49 (5.52)	−11.08 (5.60)*
Pensioner (lag)	−3.47 (2.53)	−1.73 (2.37)	−1.43 (1.81)	4.53 (5.45)	−8.91 (10.69)	−2.43 (5.00)	−6.87 (5.54)	−4.37 (7.05)
Unemployed (lag)	−3.22 (2.81)	−4.43 (1.50)**	−3.43 (3.01)	−3.24 (3.76)	−8.23 (8.98)	−11.18 (4.59)*	−6.86 (4.97)	−12.76 (6.03)*
Student (lag)	−8.12 (5.11)	−2.38 (1.76)	−4.79 (3.22)	−0.18 (4.44)	−9.35 (8.02)	0.59 (7.92)	−13.17 (6.68)*	−7.75 (7.48)
Inactive (lag)	3.93 (2.81)	−0.52 (2.01)	−0.96 (2.41)	−4.86 (4.19)	−13.44 (9.60)	−6.95 (4.33)	−2.30 (5.89)	−15.12 (6.17)*
Sick/disabled (lag)	−2.68 (3.91)	−9.43 (3.11)**	−5.03 (2.39)*	−0.54 (4.09)	0.65 (14.40)	−10.28 (4.99)*	−5.91 (4.83)	−18.82 (7.33)*
Other status (lag)	0.04 (5.45)	1.93 (3.04)	4.46 (3.97)	0.80 (5.29)	−17.01 (9.86)	−9.22 (5.40)	0.48 (9.85)	−6.45 (5.74)
Disp. income (lag)	0.00 (0.00)	0.00 (0.00)**	0.00 (0.00)***	0.00 (0.00)	0.00 (0.00)*	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Pocketbook eval. (lag)	0.02 (0.02)	0.05 (0.01)***	0.07 (0.01)***	0.06 (0.02)**	0.01 (0.03)	0.09 (0.02)***	0.05 (0.02)*	−0.02 (0.05)
Female	−0.46 (0.78)	−1.32 (0.30)***	0.13 (0.31)	0.54 (1.08)	−0.35 (1.52)	−0.08 (1.64)	1.38 (0.99)	5.74 (2.93)
Num.Obs.	7020	41 611	31 074	3000	1839	3514	2799	438
Std.Errors	Household	Household	Household	Household	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓	✓	✓	✓	✓

Models underlying the heterogeneity plot for economic status groups (LES). Columns are subgroup regressions by labor-force status category. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

K Multi-year retrospection

The main analysis measures policy-induced income changes on a year-to-year basis. However, the retrospection literature suggests that voters may evaluate governments based on cumulative experience rather than recent changes alone. This section tests two alternative retrospection windows. The first uses cumulative two-year income changes, capturing a broader evaluation horizon. The second uses the running average of income changes since the start of the current governing period (2011 for the Coalition government, 2017 for the Conservative government), reflecting full-tenure retrospection. The year 2016 is excluded from the tenure measure because the Policy Effects Tool data for that year are unavailable. If voters are more responsive to cumulative policy experiences, these specifications should show a stronger differential policy effect than the year-on-year baseline.

The results in Table K1 can be compared directly with the main year-on-year estimates reported in the body of the paper (the 1-year baseline is omitted from the table as these are identical to the main results). At the two-year horizon, the differential policy effect remains small and non-significant. At the full-tenure horizon, the differential turns negative and significant, indicating if anything that cumulated policy-induced changes receive less weight than non-policy changes. Neither horizon provides evidence that cumulative policy exposure improves attribution.

Table K1: Multi-year retrospection (asinh): 2-year and full-tenure windows

	Total	Decomposed	Total	Decomposed
Δy^{total}	0.09 (0.02)***	0.10 (0.02)***		
Δy^{policy}		−0.04 (0.03)		
Δy^{total}			0.05 (0.02)*	0.06 (0.02)**
Δy^{policy}				−0.09 (0.03)**
Con. (lag) x Coalition	65.79 (5.38)***	65.78 (5.38)***	65.38 (5.76)***	65.35 (5.76)***
Con. (lag) x Conservative	70.19 (5.40)***	70.18 (5.40)***	69.95 (5.79)***	69.93 (5.79)***
Lab. (lag) x Coalition	−10.34 (5.37)	−10.36 (5.37)	−10.32 (5.76)	−10.35 (5.76)
Lab. (lag) x Conservative	−8.78 (5.35)	−8.81 (5.35)	−8.88 (5.75)	−8.91 (5.74)
Lib Dem (lag) x Coalition	60.17 (0.85)***	60.15 (0.85)***	57.77 (0.72)***	57.73 (0.72)***
Lib Dem (lag) x Conservative	−0.68 (1.14)	−0.67 (1.14)	−1.16 (1.14)	−1.17 (1.14)
Other (lag) x Coalition	−9.87 (5.37)	−9.87 (5.38)	−9.96 (5.77)	−9.97 (5.76)
Other (lag) x Conservative	−3.44 (5.40)	−3.47 (5.40)	−3.67 (5.79)	−3.69 (5.79)
Employee (lag)	−1.72 (0.50)***	−1.70 (0.50)***	−2.39 (0.45)***	−2.34 (0.45)***
Pensioner (lag)	−1.71 (0.58)**	−1.62 (0.58)**	−2.05 (0.52)***	−1.69 (0.54)**
Unemployed (lag)	−4.91 (0.80)***	−4.90 (0.80)***	−5.59 (0.68)***	−5.48 (0.68)***
Student (lag)	−3.85 (1.01)***	−3.80 (1.01)***	−3.17 (0.86)***	−2.97 (0.86)***
Inactive (lag)	−1.67 (0.74)*	−1.67 (0.74)*	−2.82 (0.66)***	−2.78 (0.66)***
Sick/disabled (lag)	−5.10 (0.77)***	−5.03 (0.77)***	−5.31 (0.67)***	−5.15 (0.67)***
Other status (lag)	1.38 (1.58)	1.39 (1.58)	0.38 (1.49)	0.49 (1.49)
Disp. income (lag)	0.00 (0.00)**	0.00 (0.00)**	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.01)***	0.06 (0.01)***	0.06 (0.00)***	0.06 (0.00)***
Female	−0.78 (0.23)***	−0.78 (0.23)***	−0.50 (0.20)*	−0.48 (0.20)*
Age	0.07 (0.01)***	0.07 (0.01)***	0.07 (0.01)***	0.08 (0.01)***
Num.Obs.	60 701	60 701	82 165	82 165
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Multi-year retrospection using asinh-transformed income changes. The 1-year baseline is omitted; columns report 2-year and full-tenure windows, each split into total and decomposed specifications. Full tenure is a running average from government-period start (2011 for Coalition, 2017 for Conservative; 2016 excluded due to missing simulation data). The policy-induced coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

L 90% Confidence Intervals (Rainey 2014)

Following Rainey (2014), Table L1 reports 90% confidence intervals for the policy signal coefficients for the main specifications. If the 90% CI excludes a meaningful effect size, the null of negligible policy-sensitivity is supported even without a statistically significant point estimate.

Table L1: Policy-induced vs total pocketbook changes: 90% confidence intervals for policy-induced coefficient (Rainey 2014)

Specification	Estimate	90% CI Lower	90% CI Upper
Linear (£1000/mo)	-0.0314	-0.2767	0.2139
Asinh transform	0.0004	-0.0450	0.0458

Following Rainey (2014), 90% confidence intervals are reported to assess substantive significance. If the 90% CI excludes a meaningful effect size, the effect can be considered negligibly small even if not precisely zero.

M Attribution Ratio Confidence Interval

The main text reports a confidence interval for β_p/β_{np}, the ratio of voter responsiveness to policy-induced versus non-policy income changes in Equation 1. This appendix explains how the interval is derived.

The main specification includes both total income change Δy^{total} and the policy-induced component Δy^{policy} as regressors. In this parameterization, the coefficient on Δy^{total} estimates β_{np} (the baseline responsiveness to all income changes), while the coefficient on Δy^{policy} estimates the differential $\delta = \beta_p - \beta_{np}$. The ratio of interest is therefore:

$$\frac{\beta_p}{\beta_{np}} = 1 + \frac{\delta}{\beta_{np}} \tag{5}$$

One standard approach is the *delta method*, which approximates the sampling distribution of a nonlinear function of estimated parameters via a first-order Taylor expansion (see e.g. Wooldridge 2010, Ch. 3). Defining $g(\delta, \beta_{np}) = 1 + \delta/\beta_{np}$, the approximate variance is:

$$\mathrm{Var}(g) \approx \left(\frac{1}{\beta_{np}}\right)^2 \mathrm{Var}(\delta) + \left(\frac{\delta}{\beta_{np}^2}\right)^2 \mathrm{Var}(\beta_{np}) + 2\left(\frac{1}{\beta_{np}}\right)\left(\frac{-\delta}{\beta_{np}^2}\right)\mathrm{Cov}(\delta, \beta_{np}) \tag{6}$$

where the variances and covariance are taken from the cluster-robust variance–covariance matrix of the main model.

Table M1 reports delta-method confidence intervals. In the asinh specification, the point estimate is $\hat{\beta}_p/\hat{\beta}_{np} = 1.01$, with 90% and 95% CIs of [0.24, 1.77] and [0.10, 1.92], respectively. The raw £ specification yields a point estimate of 0.80 with wider intervals (90%: $[-0.67, 2.28]$; 95%: $[-0.96, 2.56]$). A ratio of 1 corresponds to Fiorina's outcome-based model of equal weighting; the point estimates are close to unity, and neither specification's confidence interval rejects it.

Table M1: Policy-induced vs total pocketbook changes: Attribution ratio ($\hat{\beta}_p/\hat{\beta}_r$) with delta-method confidence intervals

Spec.	$\hat{\beta}_r$	$\hat{\delta}$	$\hat{\beta}_p$	Ratio	SE(ratio)	DM 90%	DM 95%
Asinh (main)	0.0596	0.0004	0.0600	1.006	0.463	[0.244, 1.769]	[0.098, 1.915]
Raw (£1000/mo)	0.1581	-0.0314	0.1267	0.801	0.897	[-0.673, 2.276]	[-0.956, 2.559]

$hatbeta_r$ = coefficient on total income (base weight); *hatdelta* = differential policy effect; $hatbeta_p = hatbeta_r + hatdelta$ (implied policy weight); Ratio $= hatbeta_p/hatbeta_r = 1 + hatdelta/hatbeta_r$. Delta-method CIs use a Wald-type normal approximation. N = 91,295. Derived from the main decomposition specification.

Connecting these intervals to the simulation in Figure 1: the point estimate of approximately 1:1 is consistent with Fiorina's equal-weighting (1:1) benchmark, and the confidence intervals cannot reject it. The upper bounds of the asinh interval reach approximately 1.8:1 at 90% and 1.9:1 at 95%, ruling out the higher attribution ratios (5:1, 10:1) from the simulation but permitting modest differential weighting. At the empirically observed level of approximately 80% non-policy variation, even ratios up to 2:1 yield concordant sanctions only modestly above the no-attribution baseline, because the dominant source of income variation remains non-policy shocks that cannot be attributed to the government.

N Nominal decomposition robustness check

CPI indexation procedure

The decomposition of income changes into policy-induced and non-policy components requires a decision about how to handle routine nominal adjustments to policy parameters. Most tax-benefit systems regularly uprate monetary thresholds – such as tax brackets, benefit rates, and allowance levels – to keep pace with inflation. These inflation-matching adjustments are largely mechanical: they preserve the real value of existing policy settings rather than representing deliberate changes in generosity. To ensure that such routine adjustments are absorbed into the non-policy component, the main decomposition applies Consumer Price Index (CPI) indexation factors when computing counterfactual incomes.

Operationally, CPI indexation works as follows. When computing the counterfactual $s_0(c_{i,1})$ – i.e., applying the period-0 policy system to the individual's period-1 characteristics – the individual's monetary variables (market incomes and other nominal amounts) in period 1 are adjusted by CPI factors to period-0 nominal levels. This ensures that the period-1 incomes are nominally comparable to the period-0 policy parameters against which they are evaluated. The same CPI adjustment is applied to $s_1(c_{i,1})$ to express all quantities in real terms. As a result, the policy-induced component $\Delta y_i^{policy} = s_1(c_{i,1}) - s_0(c_{i,1})$ captures only *real* changes in policy generosity: changes in tax-benefit rules or in nominal parameters that deviate from the inflation rate. Routine inflation-matching uprating, by contrast, is netted out and absorbed into the non-policy component.

This approach has the additional advantage of capturing a specific form of "policy drift" (Hacker 2004): when governments freeze nominal benefit amounts or allow thresholds to rise more slowly than inflation ("bracket creep"; Paulus and Tasseva 2020), the CPI-indexed decomposition correctly registers these as real policy changes, since the nominal parameters have deviated from the inflation benchmark. The choice of CPI as the indexation benchmark follows Bargain and Callan (2010), who recommend it as a neutral standard. Because the analysis uses year-on-year intervals, the choice of price index is unlikely to materially affect the results.

Robustness check: no indexation

The main results decompose income changes using CPI indexation, meaning that routine inflation-matching adjustments to policy parameters (e.g., uprating benefit thresholds in line with prices) are not counted as policy changes. As a robustness check, Table N1 replicates the main specifications using a nominal decomposition that applies no CPI adjustment to the counterfactual. Under the nominal decomposition, *all* changes to nominal policy parameters – including routine inflation matching – are classified as policy-induced. This provides an upper bound on the scope of the policy signal, since it attributes more income variation to government action.

O Crude forms of attribution

Even if voters cannot identify the specific causal mechanisms behind income changes, they might engage in cruder forms of attribution based on the *source* of income changes. Instead of distinguishing income changes caused by changes in policy, they may just detect changes in the amounts they receive in transfers or pay in taxes. For instance, they might reward incumbents when their tax burden decreases or benefit payments increase, regardless of whether these changes stem from policy reforms or changes in personal circumstances (like moving tax brackets or becoming eligible for existing benefits). Such crude attribution would represent a more pragmatic, if imperfect,

Table N1: Nominal decomposition robustness: Policy-induced and non-policy pocketbook changes

	Total (raw)	Decomposed (raw)	Total (asinh)	Decomposed (asinh)
Δy^{total}	0.15 (0.03)***	0.16 (0.05)**	0.06 (0.02)**	0.06 (0.02)**
Δy^{policy}		−0.03 (0.15)		−0.01 (0.03)
Con. (lag) x Coalition	66.26 (4.79)***	66.26 (4.79)***	66.25 (4.79)***	66.25 (4.79)***
Con. (lag) x Conservative	70.71 (4.83)***	70.71 (4.83)***	70.67 (4.82)***	70.65 (4.82)***
Lab. (lag) x Coalition	−9.32 (4.79)	−9.32 (4.79)	−9.36 (4.79)	−9.36 (4.79)
Lab. (lag) x Conservative	−7.83 (4.77)	−7.84 (4.77)	−7.89 (4.77)	−7.92 (4.77)
Lib Dem (lag) x Coalition	57.77 (0.69)***	57.77 (0.69)***	57.79 (0.69)***	57.79 (0.69)***
Lib Dem (lag) x Conservative	−0.62 (1.09)	−0.62 (1.09)	−0.64 (1.09)	−0.64 (1.09)
Other (lag) x Coalition	−8.85 (4.80)	−8.85 (4.80)	−8.89 (4.80)	−8.89 (4.80)
Other (lag) x Conservative	−2.55 (4.84)	−2.56 (4.84)	−2.62 (4.83)	−2.64 (4.84)
Employee (lag)	−2.03 (0.43)***	−2.03 (0.43)***	−2.05 (0.43)***	−2.05 (0.43)***
Pensioner (lag)	−1.91 (0.49)***	−1.91 (0.50)***	−1.96 (0.49)***	−1.95 (0.50)***
Unemployed (lag)	−5.15 (0.64)***	−5.15 (0.64)***	−5.24 (0.64)***	−5.24 (0.64)***
Student (lag)	−2.94 (0.79)***	−2.93 (0.79)***	−3.01 (0.79)***	−3.00 (0.79)***
Inactive (lag)	−2.61 (0.62)***	−2.61 (0.62)***	−2.68 (0.62)***	−2.67 (0.62)***
Sick/disabled (lag)	−5.16 (0.62)***	−5.15 (0.62)***	−5.19 (0.62)***	−5.18 (0.62)***
Other status (lag)	0.04 (1.38)	0.05 (1.38)	0.00 (1.38)	0.00 (1.39)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***
Female	−0.54 (0.19)**	−0.54 (0.19)**	−0.56 (0.19)**	−0.56 (0.19)**
Age	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***
Num.Obs.	91 295	91 295	91 295	91 295
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Nominal decomposition robustness check. Unlike the main (CPI-indexed) results, the nominal decomposition treats all nominal parameter changes—including routine inflation matching—as policy-induced. Total = policy-induced + non-policy. The policy effect coefficient captures the differential effect of policy-induced vs. non-policy income changes. Outcome: incumbent support (0–100). LDV specification with wave fixed effects and household-level clustered standard errors. * p<.05, ** p<.01, *** p<.001.

Table O1: Crude attribution: Income changes by source and outlier robustness

	Asinh	Raw	Raw, winsorized 1%	Raw, trimmed p1/p99
Δy^{market}	0.05 (0.04)	0.57 (0.22)**	0.50 (0.30)	0.52 (0.37)
$\Delta y^{transfers}$	0.05 (0.03)	0.10 (0.16)	−0.15 (0.40)	0.77 (0.52)
Δy^{taxes}	−0.04 (0.05)	1.11 (0.48)*	0.43 (0.84)	−0.43 (1.18)
Con. (lag) x Coalition	66.28 (4.79)***	66.26 (4.79)***	66.26 (4.80)***	65.83 (4.79)***
Con. (lag) x Conservative	70.52 (4.83)***	70.51 (4.83)***	70.51 (4.83)***	70.64 (4.82)***
Lab. (lag) x Coalition	−9.33 (4.79)	−9.34 (4.79)	−9.35 (4.79)	−9.30 (4.78)
Lab. (lag) x Conservative	−8.05 (4.77)	−8.06 (4.77)	−8.08 (4.77)	−7.78 (4.76)
Lib Dem (lag) x Coalition	57.79 (0.69)***	57.80 (0.69)***	57.80 (0.69)***	57.50 (0.71)***
Lib Dem (lag) x Conservative	−0.64 (1.09)	−0.61 (1.09)	−0.62 (1.09)	−0.69 (1.11)
Other (lag) x Coalition	−8.86 (4.80)	−8.87 (4.80)	−8.88 (4.80)	−8.94 (4.79)
Other (lag) x Conservative	−2.77 (4.84)	−2.79 (4.84)	−2.80 (4.84)	−2.57 (4.83)
Employee (lag)	−2.07 (0.43)***	−2.04 (0.43)***	−2.04 (0.43)***	−2.28 (0.44)***
Pensioner (lag)	−1.94 (0.49)***	−1.87 (0.50)***	−1.90 (0.50)***	−1.47 (0.52)**
Unemployed (lag)	−5.20 (0.64)***	−5.18 (0.64)***	−5.22 (0.64)***	−4.62 (0.66)***
Student (lag)	−2.97 (0.79)***	−2.95 (0.79)***	−2.97 (0.79)***	−2.09 (0.81)**
Inactive (lag)	−2.65 (0.62)***	−2.60 (0.62)***	−2.63 (0.62)***	−2.26 (0.64)***
Sick/disabled (lag)	−5.21 (0.62)***	−5.14 (0.62)***	−5.16 (0.62)***	−4.78 (0.64)***
Other status (lag)	−0.01 (1.39)	0.05 (1.39)	0.01 (1.39)	0.55 (1.44)
Disp. income (lag)	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***	0.00 (0.00)***
Pocketbook eval. (lag)	0.06 (0.00)***	0.06 (0.00)***	0.06 (0.00)***	0.05 (0.00)***
Female	−0.55 (0.19)**	−0.54 (0.19)**	−0.55 (0.19)**	−0.07 (0.20)
Age	0.08 (0.01)***	0.08 (0.01)***	0.08 (0.01)***	0.07 (0.01)***
Num.Obs.	91 295	91 295	91 295	86 816
Std.Errors	Household	Household	Household	Household
FE: wave	✓	✓	✓	✓
Lagged DV	✓	✓	✓	✓

Consolidated crude-attribution table. Column (1) reports baseline source decomposition in asinh form. Columns (2)-(4) report raw-scale estimates under baseline, winsorized (1st/99th percentile), and trimmed (drop outside p1/p99) outlier treatments; the baseline raw specification appears once. All columns use the same LDV specification with wave fixed effects, lagged party-support interactions with government dummies, and household-level clustered standard errors. Outcome: incumbent support (0–100). * p<.05, ** p<.01, *** p<.001.

approach to disciplining incumbents and would suggest at least some recognition of government involvement in their finances.

If voters engage in such crude attribution driven by the source of income change rather than its cause, we would expect voters to respond more strongly to income changes due to changes in what they pay in taxes (Δy^{taxes}) or receive in transfers ($\Delta y^{transfers}$) compared to changes in market incomes (Δy^{market}). To examine this, I estimate models of incumbent support on income changes decomposed into these three types, using the main specification from before. Because the three source components sum to total income and are correlated, the individual coefficients are less precisely estimated than the single total-income coefficient reported elsewhere; the comparison of interest is the relative magnitude across sources within each specification rather than the absolute coefficient levels.[17]

[17]Note that to facilitate interpretation, the coefficients on 'taxes' are flipped such that a positive unit change is an increase in disposable income due to taxes (i.e. a *reduction* in taxes paid).

Table O1 shows the results. As shown, there is little evidence of even crude attribution. Voters respond to market income changes while showing no systematically stronger response to tax or transfer-related changes. The coefficients on tax and transfer changes are neither consistently larger than the market income coefficient nor robustly statistically significant across specifications. This is consistent with a relatively even responsiveness to income changes regardless of their source. Thus, voters treat a pay rise and a tax cut equivalently when evaluating incumbents with seemingly little recognition of the government's differential role in these income sources.

The one apparent exception to this pattern is the tax coefficient in the raw (£) specification, which is nominally significant. However, the raw and inverse hyperbolic sine (asinh) functional forms yield qualitatively different results for this coefficient: significant and positive in raw £, but small and insignificant under the asinh transformation. This discrepancy is driven by a small number of extreme values in the tails of the raw income change distributions. The raw linear specification gives equal weight per pound to all observations, so individuals with very large absolute income changes exert disproportionate leverage on the estimated coefficients. The asinh transformation, which approximates the log for large values and is approximately linear near zero, compresses this tail variation and thus downweights extreme observations. To adjudicate between these results, the final two columns of Table O1 re-estimate the crude attribution model using two outlier treatments on the raw £ variables: (1) winsorization, which caps extreme values at specified percentiles rather than discarding them, and (2) trimming, which drops observations falling beyond the 1st or 99th percentile on any of the three income variables. The significant tax coefficient in the raw specification does not survive any form of outlier treatment. Under winsorization and trimming, the tax coefficient shrinks substantially and loses statistical significance, converging toward the asinh estimates. The market income coefficient is more stable across outlier treatments; the transfer coefficient, while never significant, is also sensitive to outlier handling. This confirms that the raw–asinh discrepancy is driven by outlier leverage rather than a genuine difference in the relationship between tax changes and incumbent support.

P Research Ethics

Human participants. This study is based entirely on secondary analysis of the UK Household Longitudinal Study (Understanding Society), a de-identified survey dataset collected and managed by the University of Essex. The author had no direct contact with human participants. The original survey data were collected under ethical oversight by the University of Essex; no additional data collection, experimental interventions, or interactions with human participants were conducted for this study. The study is exempt from ethics review under the University of Copenhagen's institutional guidelines for secondary analysis of anonymized data. No individual respondents can be identified from the results reported.

Data access. Access to the data was granted under a Special Licence from the UK Data Service (project number 258016). The licence requires a formal application, institutional approval, and adherence to an end-user agreement governing data security and respondent confidentiality. The UKMOD-linked input data (Study Number 9147) were accessed and stored on encrypted institutional infrastructure at the University of Copenhagen in accordance with the licence conditions.

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Conflicts of interest. The author declares no conflicts of interest.

Data availability. The data used in this study are subject to the UK Data Service Special Licence and cannot be made publicly available. Qualified researchers may apply for access through the UK Data Service (https://ukdataservice.ac.uk). Replication code will be made available on the APSR Dataverse upon publication.